



Mahidol University
Wisdom of the Land

Pharmacology of Antidiabetic Drugs

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เอกสารประกอบการสอนรายวิชาเภสัชวิทยา 2 (ภกปว 305) สำหรับนักศึกษาเภสัชศาสตร์ชั้นปีที่ 3

บรรยายวันที่ 11 มีนาคม 2569 (9.00-12.00 น.) ห้อง 203 ชั้น 2 อาคารเทพรัตน์

จัดทำวันที่ 1 กุมภาพันธ์ 2564 และปรับปรุงวันที่ 2 มีนาคม 2569



Learning Objectives

After the class, students should be able to

1. Explain mechanism of actions (MOA) and pharmacodynamics of antidiabetic drugs
2. Explain pharmacokinetic properties of antidiabetic drugs
3. Explain ADRs and drug interactions of antidiabetic drugs



Criteria for diagnosis of diabetes mellitus

Polyuria and polydipsia, fatigue, and often abrupt and significant weight loss

- Symptoms of diabetes plus random blood glucose concentration ≥ 11.1 mM (200 mg/dL)^a or
- Fasting plasma glucose ≥ 7.0 mM (126 mg/dL)^b or
- Two-hour plasma glucose ≥ 11.1 mM (200 mg/dL) during an oral glucose tolerance test^c
- HbA_{1c} $\geq 6.5\%$

Note: In the absence of unequivocal hyperglycemia and acute metabolic decompensation, these criteria should be confirmed by repeat testing on a different day.

^aRandom is defined as without regard to time since the last meal.

^bFasting is defined as no caloric intake for at least 8 h.

^cThe test should be performed using a glucose load containing the equivalent of 75 g anhydrous glucose dissolved in water; this test is not recommended for routine clinical use.

Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 13e.

HbA_{1c} (or A1c) = exposure of proteins to elevated glucose produces nonenzymatic glycation of proteins, including Hb

- Represents a measure of the average glucose concentration to which the Hb has been exposed
- Reflects glycemic control over the prior 3 months



Different forms of diabetes mellitus

I. Type 1 diabetes β -cell destruction, usually leading to absolute insulin deficiency

- A. Immune mediated
- B. Idiopathic

II. Type 2 diabetes (may range from predominantly insulin resistance with relative insulin deficiency to a predominantly insulin secretory defect with insulin resistance)

Insulin sensitivity is measured as the amount of glucose cleared from the blood in response to a fixed dose of insulin

Insulin resistance = the failure of normal amounts of insulin to elicit the expected response

- Muscle and fat: a decrease in uptake of glucose from the circulation
- Liver: a blunted ability of insulin to suppress glucose production
- Adipocytes: also increased rates of lipolysis and release of fatty acids into the circulation

III. Other specific types of diabetes

- A. Monogenic disorders of β -cell function
 1. HNF-4 α (MODY 1)
 2. Glucokinase (MODY 2)
 3. HNF-1 α (MODY 3)
 4. Other forms of MODY: insulin promoter factor 1, HNF-1 β , NeuroD1, and others
 5. Permanent neonatal diabetes KCNJ11 gene encoding Kir6.2 subunit of β -cell K_{ATP} channel, insulin gene
 6. Mitochondrial DNA
- B. Genetic defects in insulin action, including type A insulin resistance, leprechaunism, Rabson-Mendenhall syndrome, lipodystrophy syndromes
- C. Diseases of the exocrine pancreas—pancreatitis, pancreatectomy, neoplasia, cystic fibrosis, hemochromatosis, fibrocalculous pancreatopathy, mutations in carboxyl ester lipase
- D. Endocrinopathies—acromegaly, Cushing syndrome, glucagonoma, pheochromocytoma, hyperthyroidism, somatostatinoma, aldosteronoma
- E. Drug or chemical induced—pyrinuron (a rodenticide no longer sold in the U.S.); see drugs listed in Table 47-3
- F. Infections—congenital rubella, cytomegalovirus
- G. Uncommon forms of immune-mediated diabetes—"stiff-person" syndrome, anti-insulin receptor antibodies
- H. Other genetic syndromes sometimes associated with diabetes—Wolfram, Down, Klinefelter, Laurence-Moon-Biedl, Prader-Willi, and Turner syndromes; Friedreich ataxia; Huntington disease; myotonic dystrophy; porphyria

IV. Gestational diabetes mellitus



Antidiabetic drugs

(Hypoglycemic or Glucose-lowering drugs)

1. Insulin analogues

2. Insulin secretagogues

- Glucagon-like peptide-1 (GLP-1) receptor agonists
- Dual GIP/GLP-1 receptor agonist
- Dipeptidylpeptidase-4 (DPP-4) inhibitors
- Sulfonylureas
- Non-sulfonylureas (Meglitinides)

3. Insulin sensitizers

- Biguanides
- Thiazolidinediones

4. Inhibition of glucose absorption

- α -glucosidase inhibitors

5. Increase of glucose excretion

- Sodium-glucose cotransporter 2 (SGLT2) inhibitors

6. Amylin analogues

7. Bile Acid Sequestrant: Colesevelam

8. Dopamine Agonist: Bromocriptine

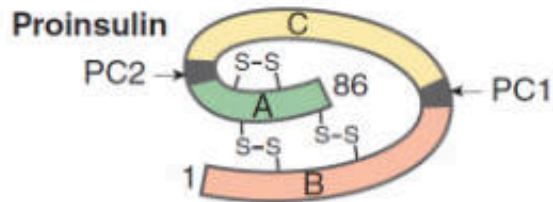
Insulin-independent

Insulin: Synthesis and post-translational processing

Preproinsulin



- SP cleavage
- Folding
- S-S bond formation



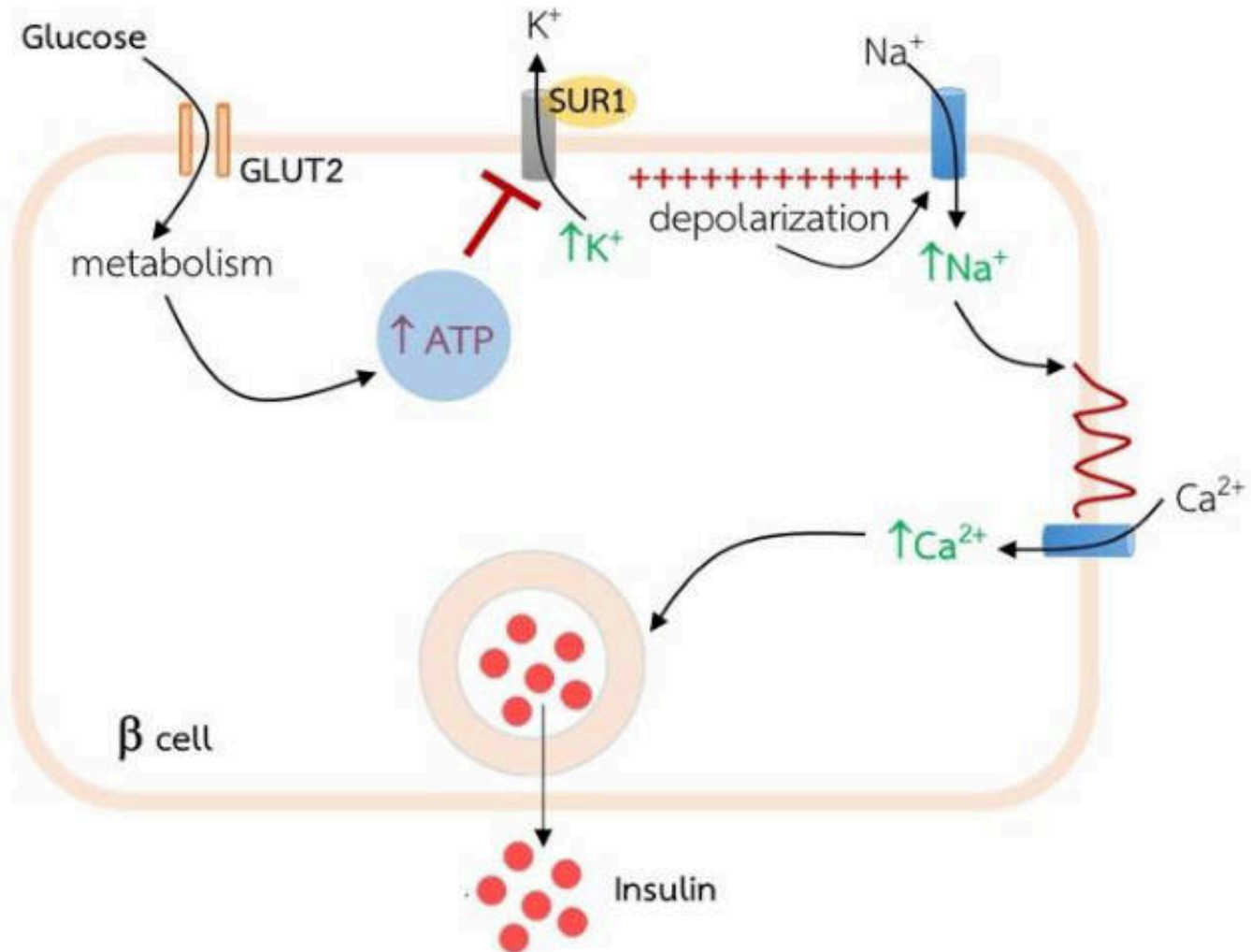
PC1: cleavage of Arg³¹/Arg³²
PC2: cleavage of Lys⁶⁴/Arg⁶⁵

Insulin



- Equimolar quantities of insulin and C-peptide are co-secreted
- **Insulin has a $t_{1/2}$ of 5–6 min** due to extensive hepatic clearance

Activation of insulin secretion by glucose



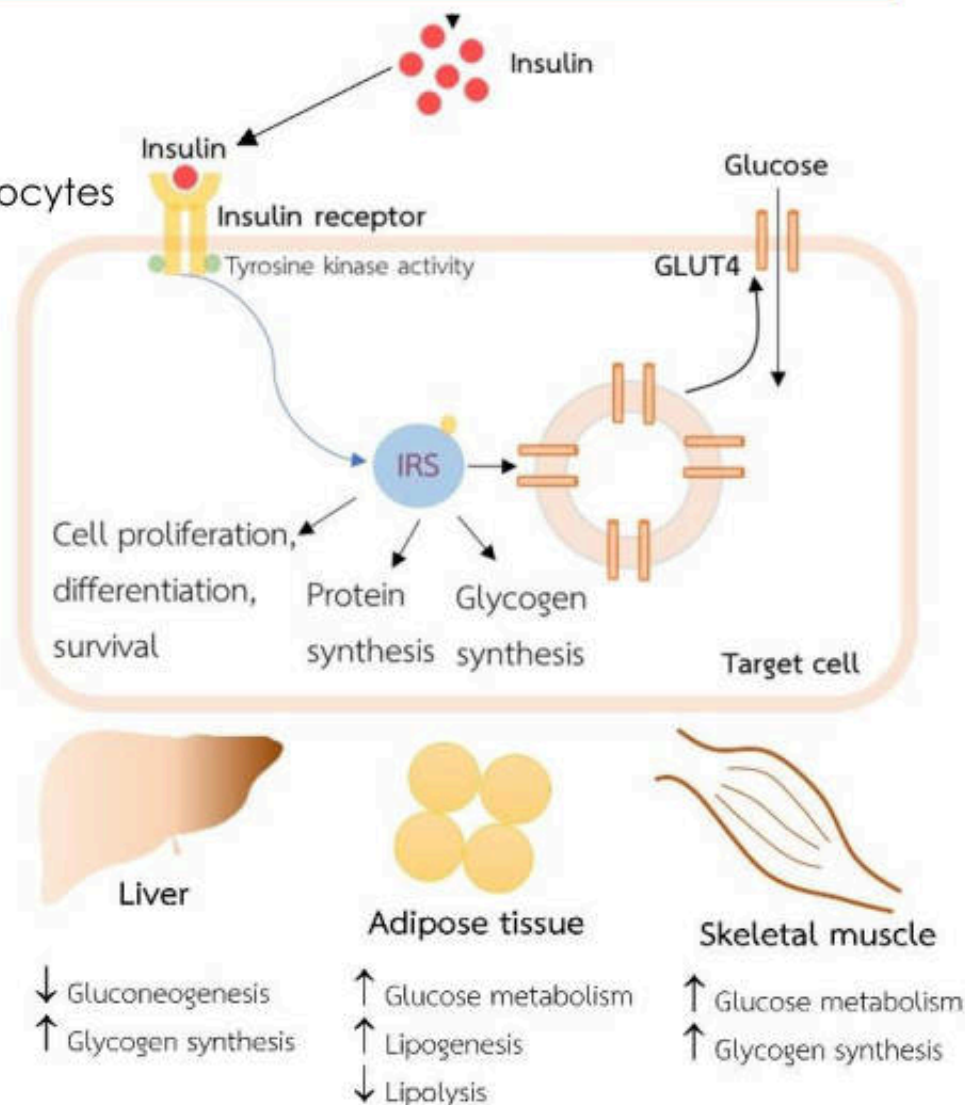


Pathways of insulin signaling

The number of receptors varies

- 40/cell on erythrocytes
- 300,000/cell on adipocytes and hepatocytes

IRS: insulin receptor substrate





Time-action profiles of insulin preparations

TYPE	PREPARATION	TIMES		
		ONSET (h)	PEAK (h)	EFFECTIVE DURATION (h)
Short acting				
	Aspart	} <0.25	} 0.5–1.5	} 3–4
	Glulisine			
	Lispro			
	Regular	0.5–1.0	2–3	4–6
Long acting				
	Detemir	1–4	0	12–20
	Glargine ^b	1–4	0 ^a	12–24
	Degludec	1–4	0	24–42
Intermediate acting →	NPH	1–2	6–10	10–16
Insulin combinations				
	Mixture: short acting (25%–50%) and long acting (50%–76%)	<0.25–1.0	1.5 ^c	Up to 10–16
Inhaled insulin				
	Afrezza	<0.25	0.5–1.5	2–3

^aGlargine, degludec, and detemir have minimal peak activity at steady-state.

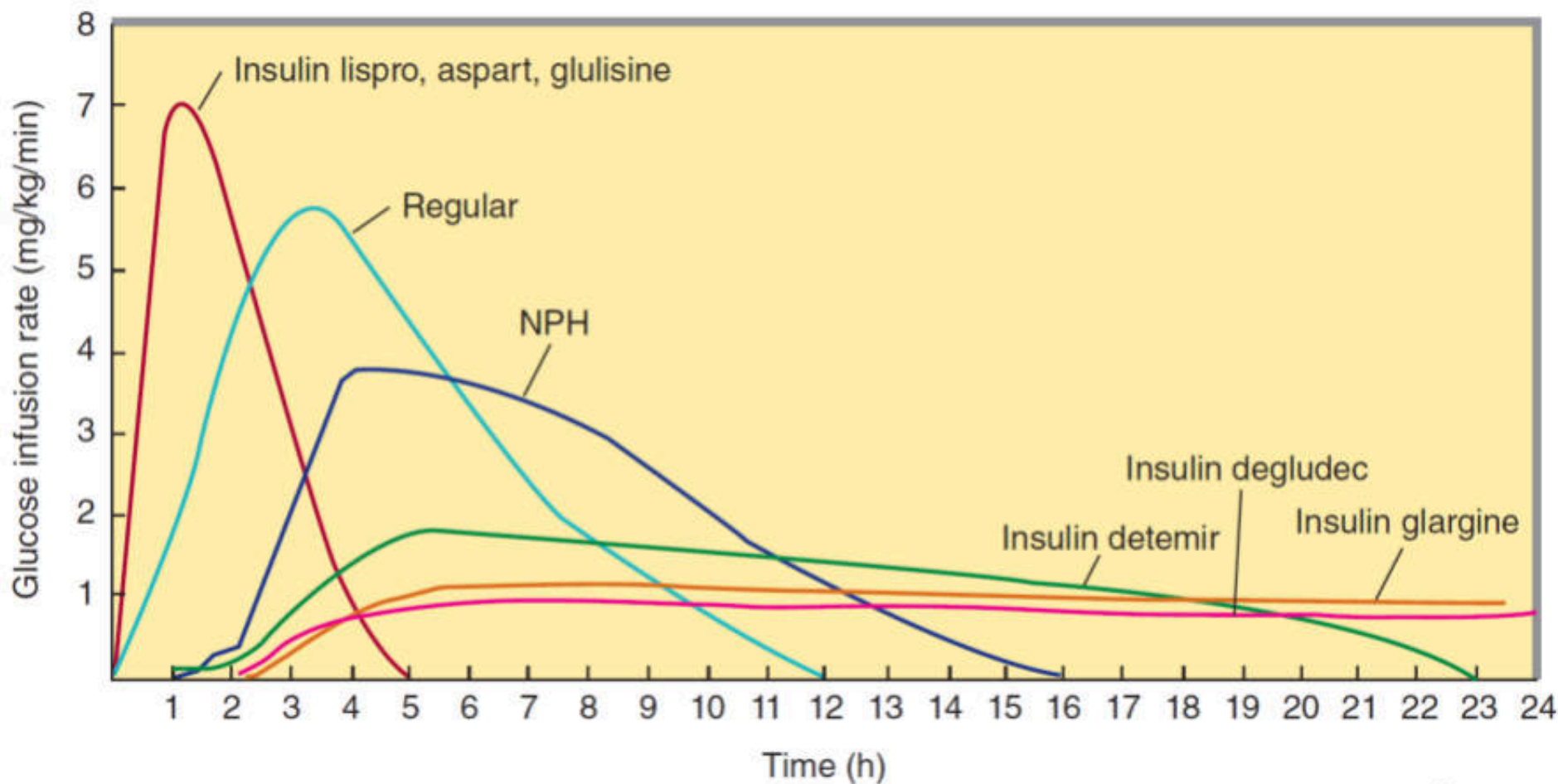
^bAvailable as a U-100 and U-300 preparation.

^cSome mixtures will have dual peaks, one at 2–3 h, the second one several hours later.

^dDiscontinued by manufacturer in the US.

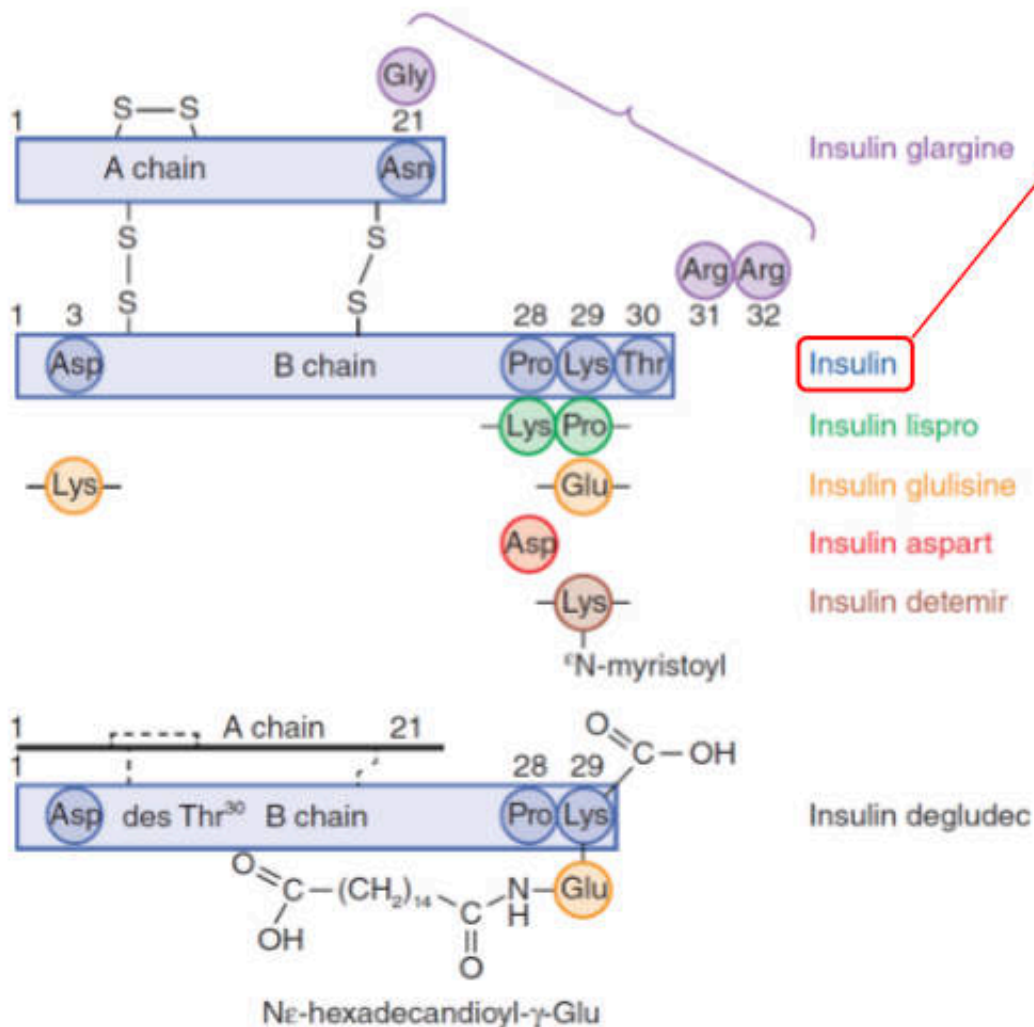


Extent and duration of action of various types of insulin





Structure of insulin analogues



- **Native or regular insulin molecules associate as hexamers in aqueous solution at a neutral pH**

- Slows absorption following s.c.

- **NPH (Neutral Protamine Hagedorn) insulin or insulin isophane**

- A suspension of native insulin complexed with zinc and protamine in a phosphate buffer
 - Cloudy or whitish solution in contrast to the clear appearance of other insulin solutions
 - Dissolves more gradually after s.c.

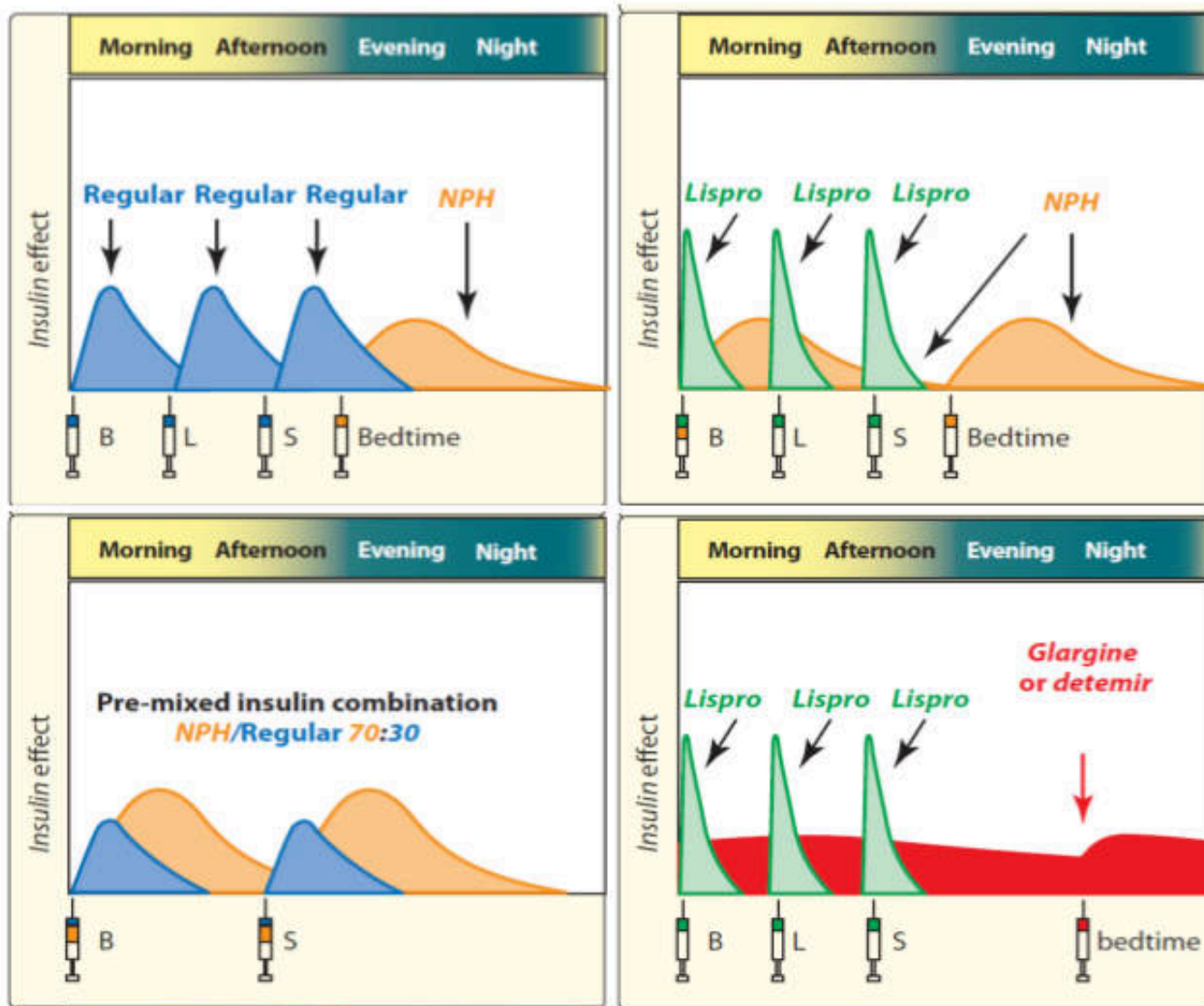


Commonly used insulin regimens

- Long-acting insulin as basal and a short-acting insulin to provide postprandial (bolus) needs
 - The basal dose is usually 40%–50% of the total daily dose, with the remainder as prandial or premeal insulin
 - The insulin dose at mealtime should reflect the anticipated carbohydrate intake
 - A supplemental scale of short-acting insulin is added to the prandial insulin dose to allow correction of the blood glucose



Examples of insulin regimen



B = breakfast; L = lunch; S = supper



Factors that affect insulin absorption

- The site of injection
 - Abdomen is the preferred in the morning because insulin is absorbed 20%–30% faster than from the arm
- The type of insulin
- Subcutaneous blood flow
 - Increased subcutaneous blood flow (brought about by massage, hot baths, or exercise) increases the rate of absorption
- Smoking
- Regional muscular activity at the site of the injection
- The volume and concentration of the injected insulin
- Depth of injection
 - More rapid onset of action if delivered IM rather than s.c.



Adverse events of insulin

- **Hypoglycemia is the major risk**
 - β -blockers have the potential for masking symptoms of hypoglycemia
- **Modest weight gain**
- **Allergic reactions** to recombinant human insulin may rarely occur
 - A reaction to the small amounts of aggregated or denatured insulin
 - Minor contaminants
 - Other components (protamine, Zn^{2+} , etc.)
- Atrophy of subcutaneous fat at the site of insulin injection (**lipoatrophy**) was a rare side effect of older insulin preparations
- **Lipohypertrophy** (enlargement of subcutaneous fat depots) has been ascribed to the lipogenic action of high local concentrations of insulin

Rotation of injection
sites is recommended



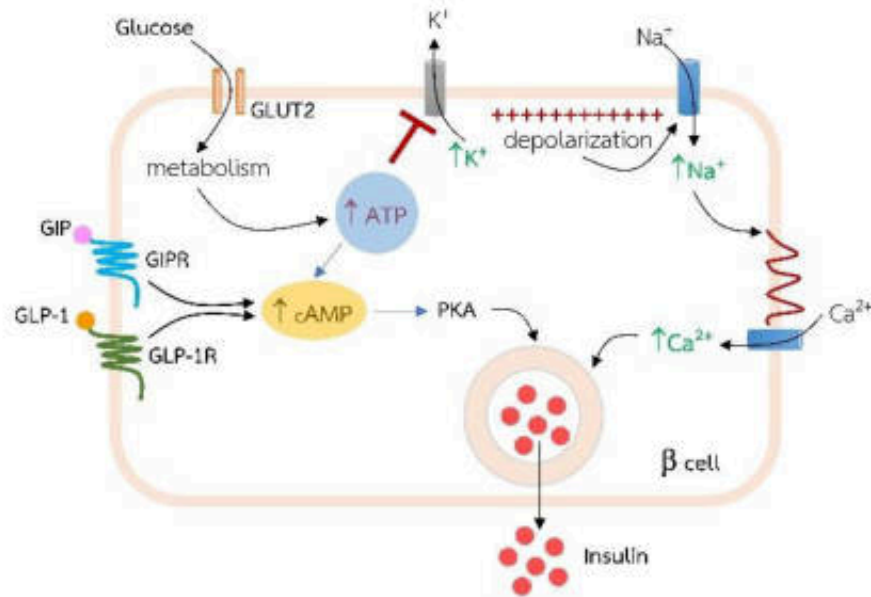
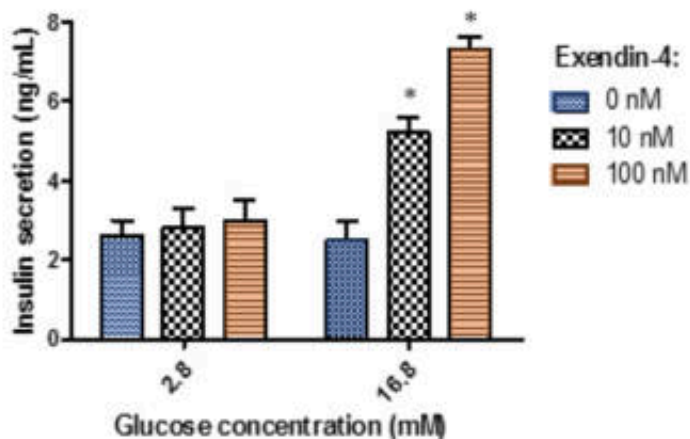
Insulin Secretagogues

1. Glucagon-like peptide-1 (GLP-1) receptor agonists
2. Dual GIP/GLP-1 receptor agonist
3. Dipeptidylpeptidase-4 (DPP-4) inhibitors
4. Sulfonylureas
5. Non-sulfonylureas (Meglitinides)

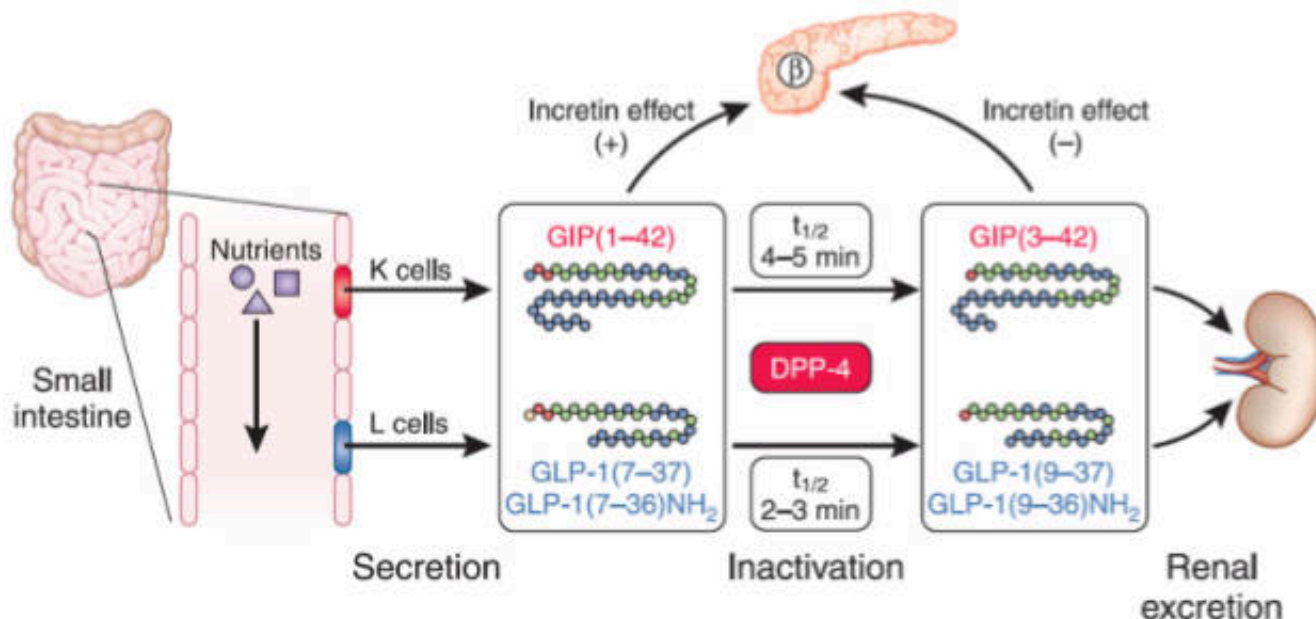
GLP-1 and GIP

- Incretins are GI hormones that are released after meals and stimulate insulin secretion
- The best-known incretins are
 - GLP-1 (glucagon-like peptide-1)
 - GIP (glucose-dependent insulintropic polypeptide or gastric inhibitory peptide)

in vitro insulin secretion



DPP-4-mediated GLP-1 degradation



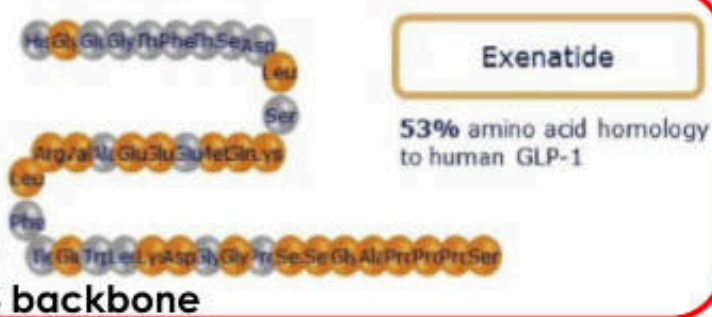
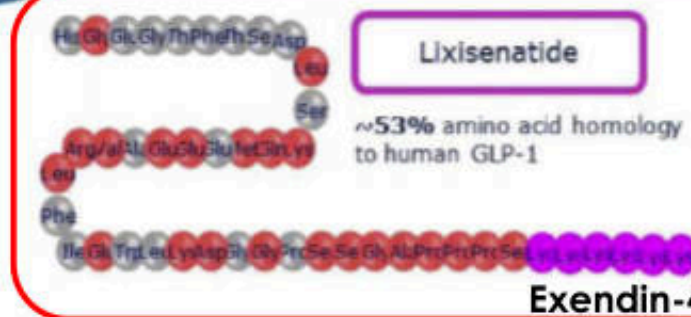
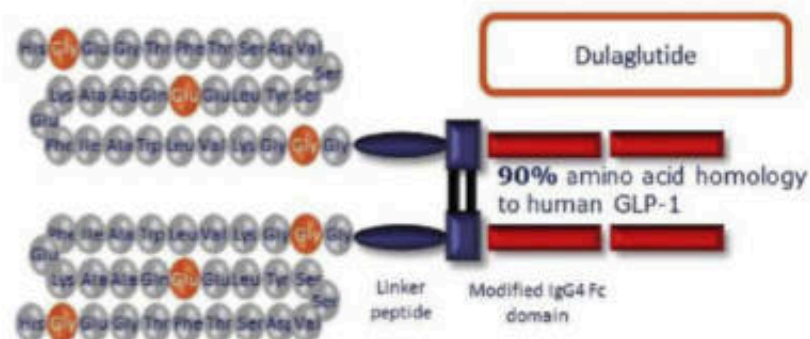
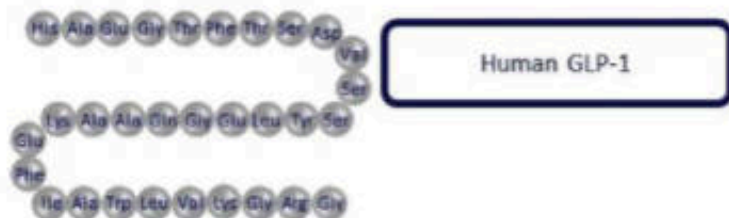
J Diabetes Invest. 2010. doi: 10.1111/j.2040-1124.2010.00022.x.

- DPP-4 is expressed as an ectoenzyme on vascular endothelial cells, on the surface of T lymphocytes, and in a circulating form
- Several agents provide nearly complete and long-lasting inhibition of DPP-4, thereby increasing the proportion of active GLP-1 from 10% to 20% of total circulating GLP-1 immunoreactivity to nearly 100%

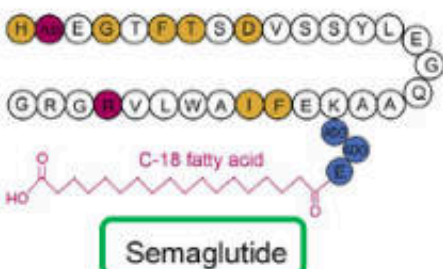


GLP-1 receptor agonists

- Modify structure to prevent degradation by DPP-4



Exendin-4 backbone



Pharmacological properties of GLP-1 receptor agonists

Parameter	Short-acting GLP-1 receptor agonist	Long-acting GLP-1 receptor agonist
Drugs	Exenatide → BID Lixisenatide → OD	Liraglutide → OD Albiglutide Dulaglutide Exenatide-LAR Semaglutide } QW
Half-life	2 to 5 h	12 h to 5-7 days
Fasting insulin secretion	Modest stimulation	Strong stimulation
Postprandial insulin secretion	Reduction	Modest stimulation
Gastric emptying rate	Deceleration	No effect
Postprandial hyperglycemia	Strong reduction	Modest reduction
Fasting blood glucose levels	Modest reduction	Strong reduction
Weight reduction	← 1-4 kg →	

Oral Semaglutide

● SNAC

Sodium N-(8-[2-hydroxybenzoyl] Amino) Caprylate



● Semaglutide



Oral semaglutide is co-formulated with SNAC.

SNAC neutralizes the pH of the gastric fluid and protects semaglutide from proteolytic degradation.

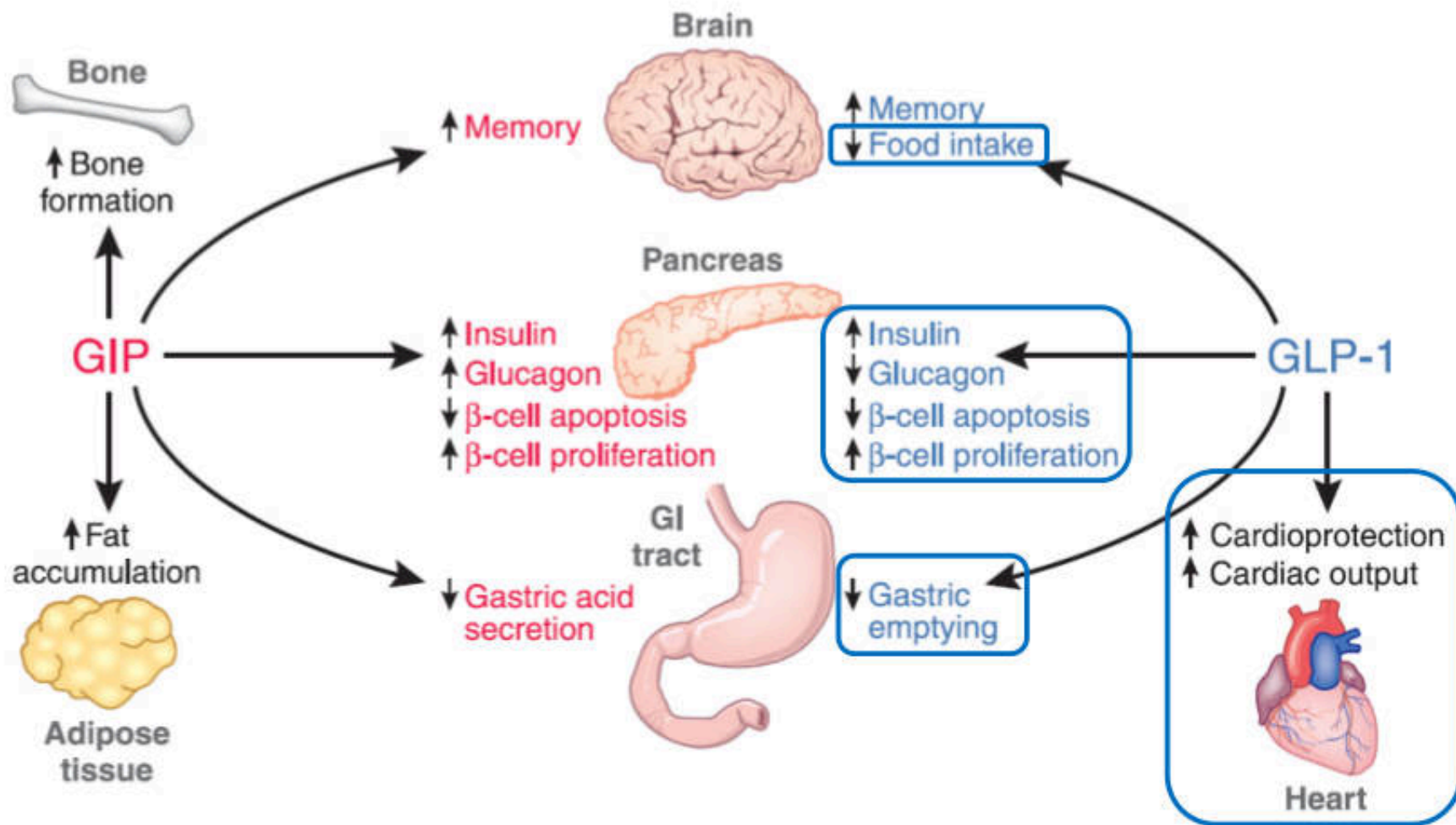
SNAC facilitates the transcellular absorption of semaglutide across the gastric mucosa.

Blood circulation

Int. J. Mol. Sci. 2021; 22: 9936. doi: 10.3390/ijms22189936

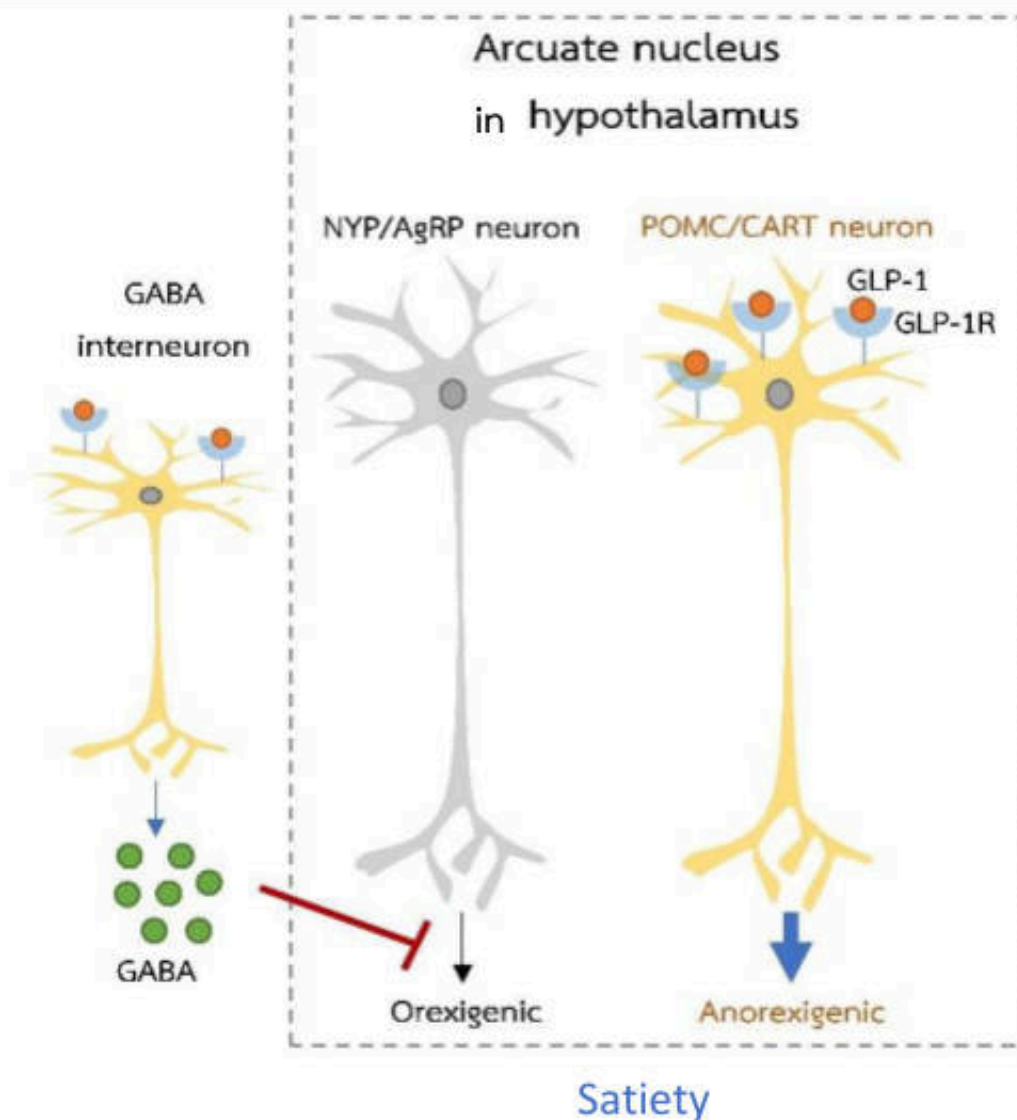
Parameter	Oral semaglutide	Subcutaneous semaglutide
AUC (nmol × h/L)	284	3026
C _{max} (nM)	15	10
T _{max} (h)	1	66
T _{1/2} (h)	152	168

Physiological effects of GLP-1 and GIP

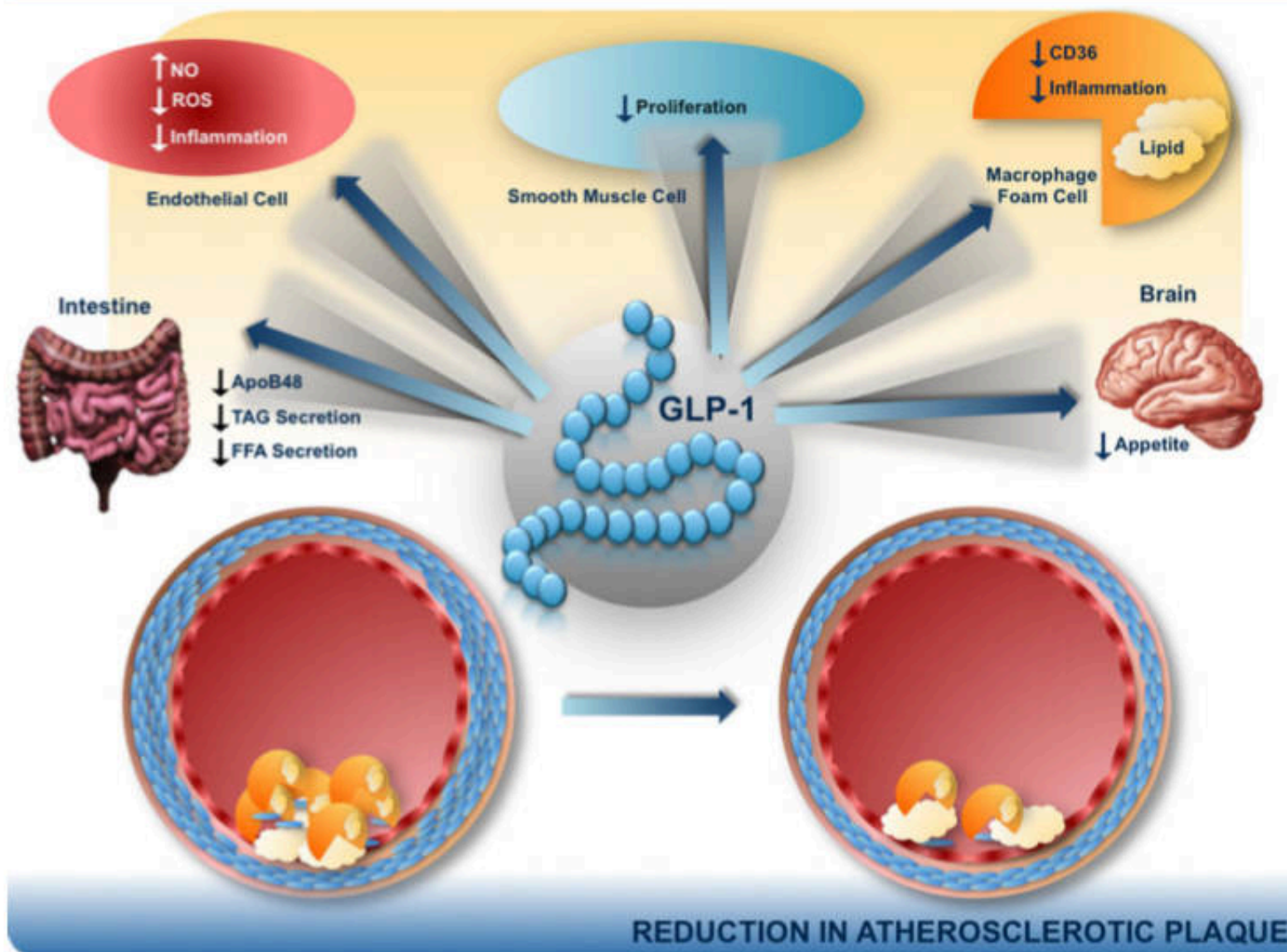


The insulinotropic effect of GLP-1 is glucose dependent

Potential mechanisms for appetite suppression of GLP-1

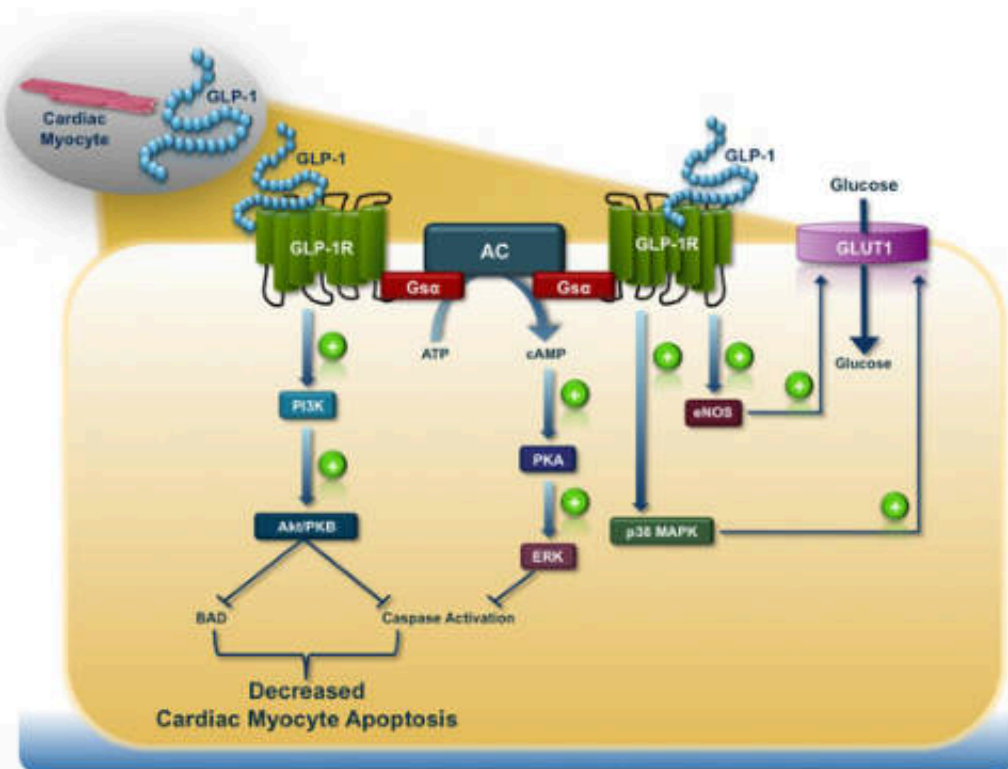


Cardiovascular benefits of incretin-based therapy

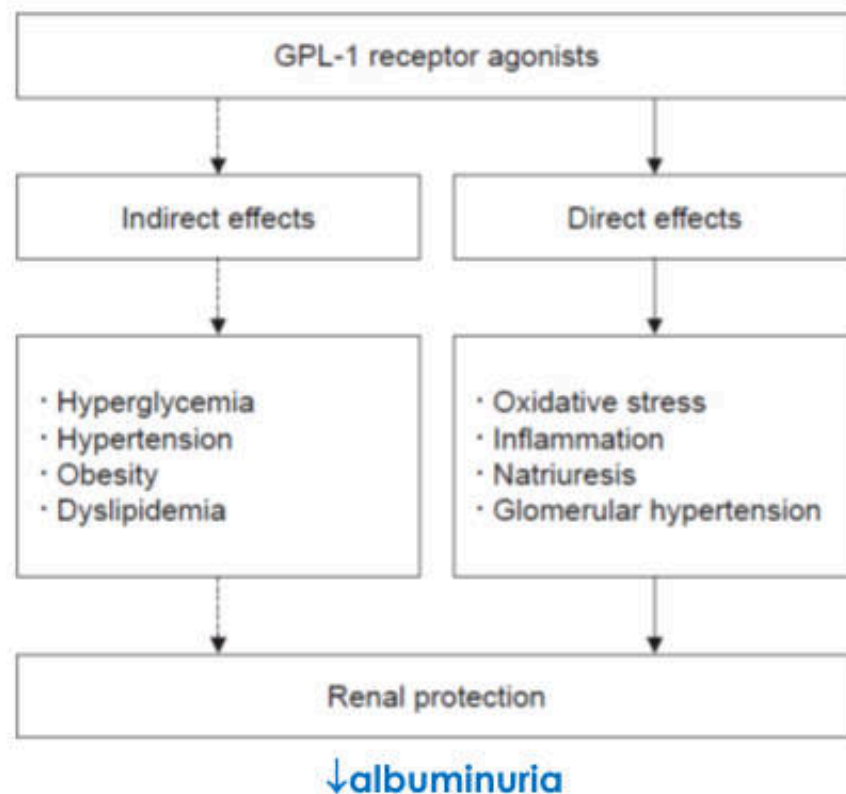


Cardiac and renal benefits of incretin-based therapy

Direct cardiac benefits



Renal benefits





Dual GIP/GLP-1 receptor agonist

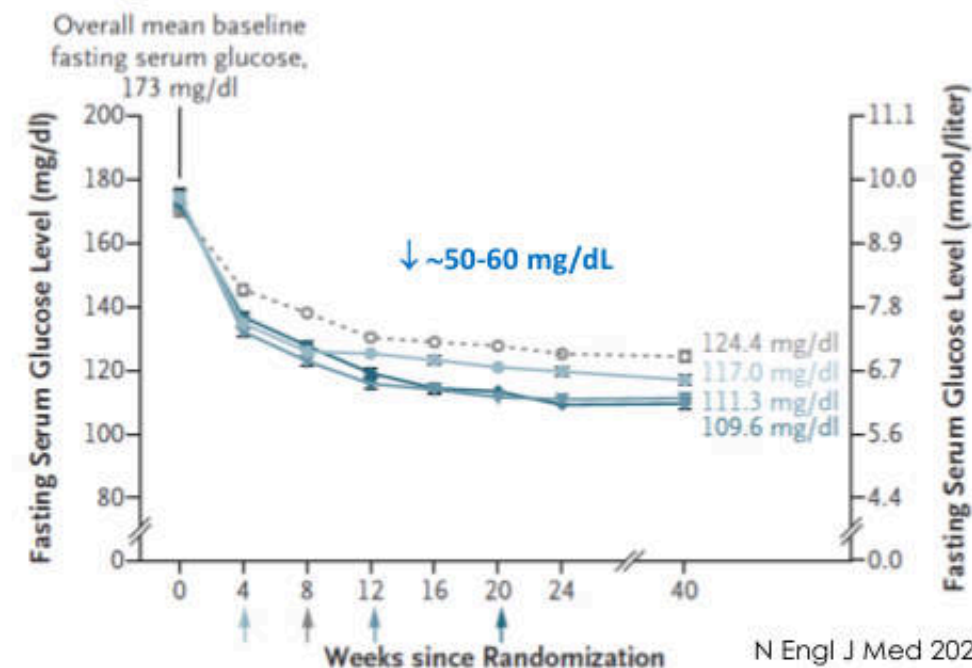
Peptide	GIPR (pIC ₅₀ ± S.E.M.)	GLP-1R (pIC ₅₀ ± S.E.M.)	GCGR (pIC ₅₀ ± S.E.M.)
GIP	7.88 ± 0.04	N.B.	N.B.
GLP-1	N.B.	8.11 ± 0.10	N.B.
GCG	N.B.	6.15 ± 2.37	6.67 ± 0.18
Semaglutide	N.B.	8.34 ± 0.17	N.B.
Tirzepatide	7.50 ± 0.10	8.09 ± 0.16	N.B.
Peptide 20	8.18 ± 0.13	8.47 ± 0.23	6.97 ± 0.18
Albiglutide	N.B.	8.42 ± 0.09	N.B.
Liraglutide	N.B.	8.50 ± 0.11	N.B.
Lixisenatide	N.B.	8.80 ± 0.13	N.B.

N.B., no binding was detected.

Nat Commun. 2022;13(1):1057. doi: 10.1038/s41467-022-28683-0

Tirzepatide

D Fasting Serum Glucose Levels

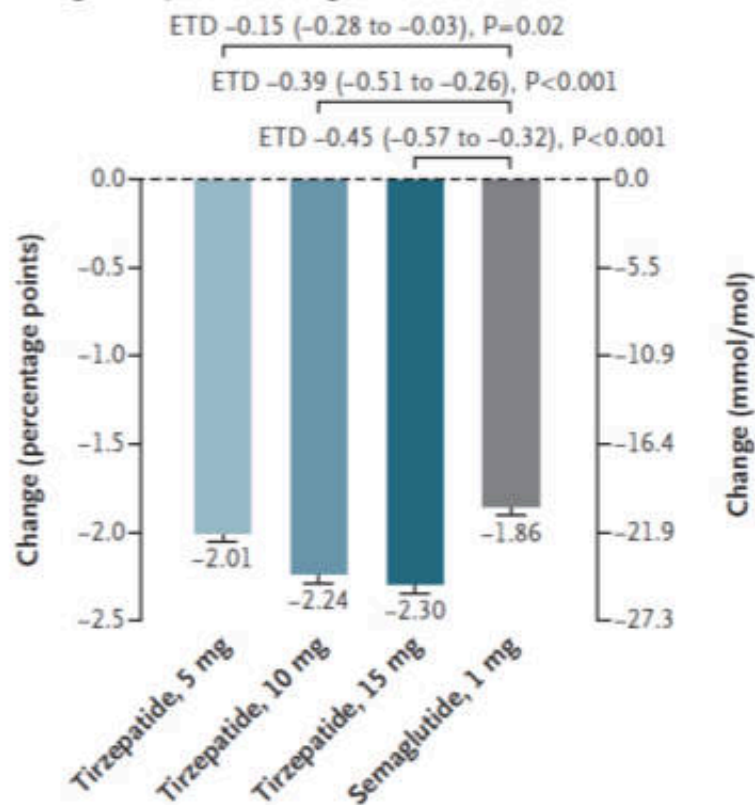




Pharmacological properties of tirzepatide

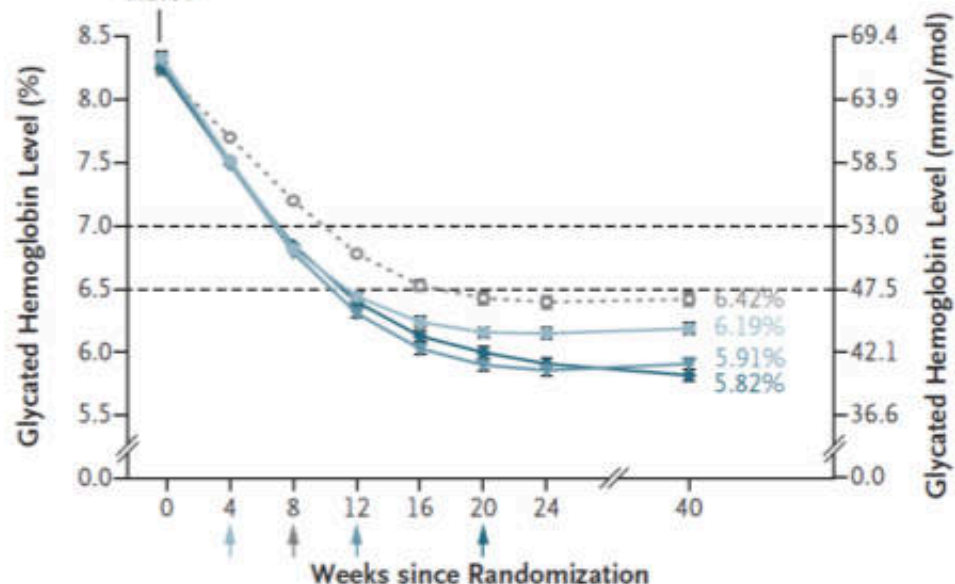
● Tirzepatide, 5 mg ▣ Tirzepatide, 10 mg ◆ Tirzepatide, 15 mg ○ Semaglutide, 1 mg

A Change in Glycated Hemoglobin Levels from Baseline



B Glycated Hemoglobin Level

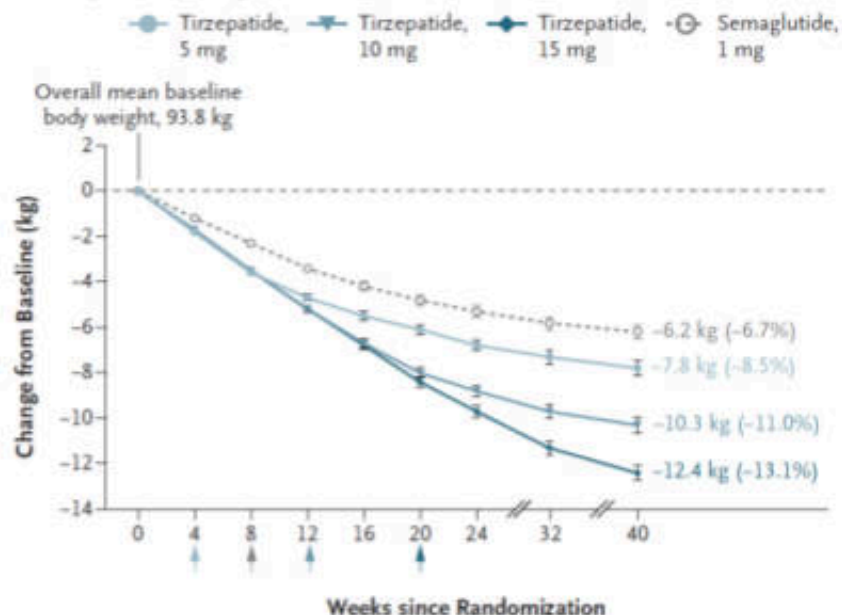
Overall mean baseline
glycated hemoglobin,
8.28%



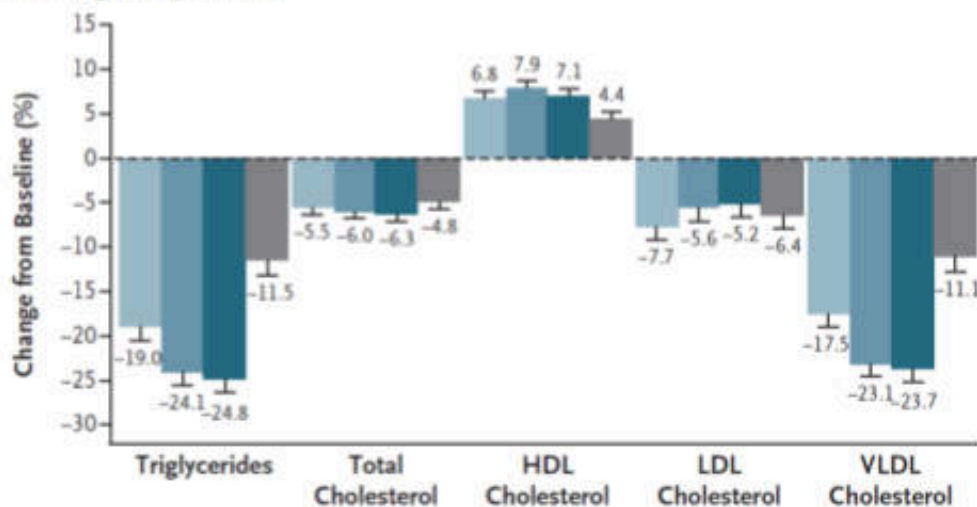


Pharmacological properties of tirzepatide (Cont.)

B Change in Body Weight from Wk 0 to Wk 40



D Change in Lipid Levels



N Engl J Med 2021;385:503-15.

- Weekly s.c. injection
- Pharmacokinetics
 - T_{max}: 1–2 days of administration
 - Half-life: 116.7 h (~5 days)

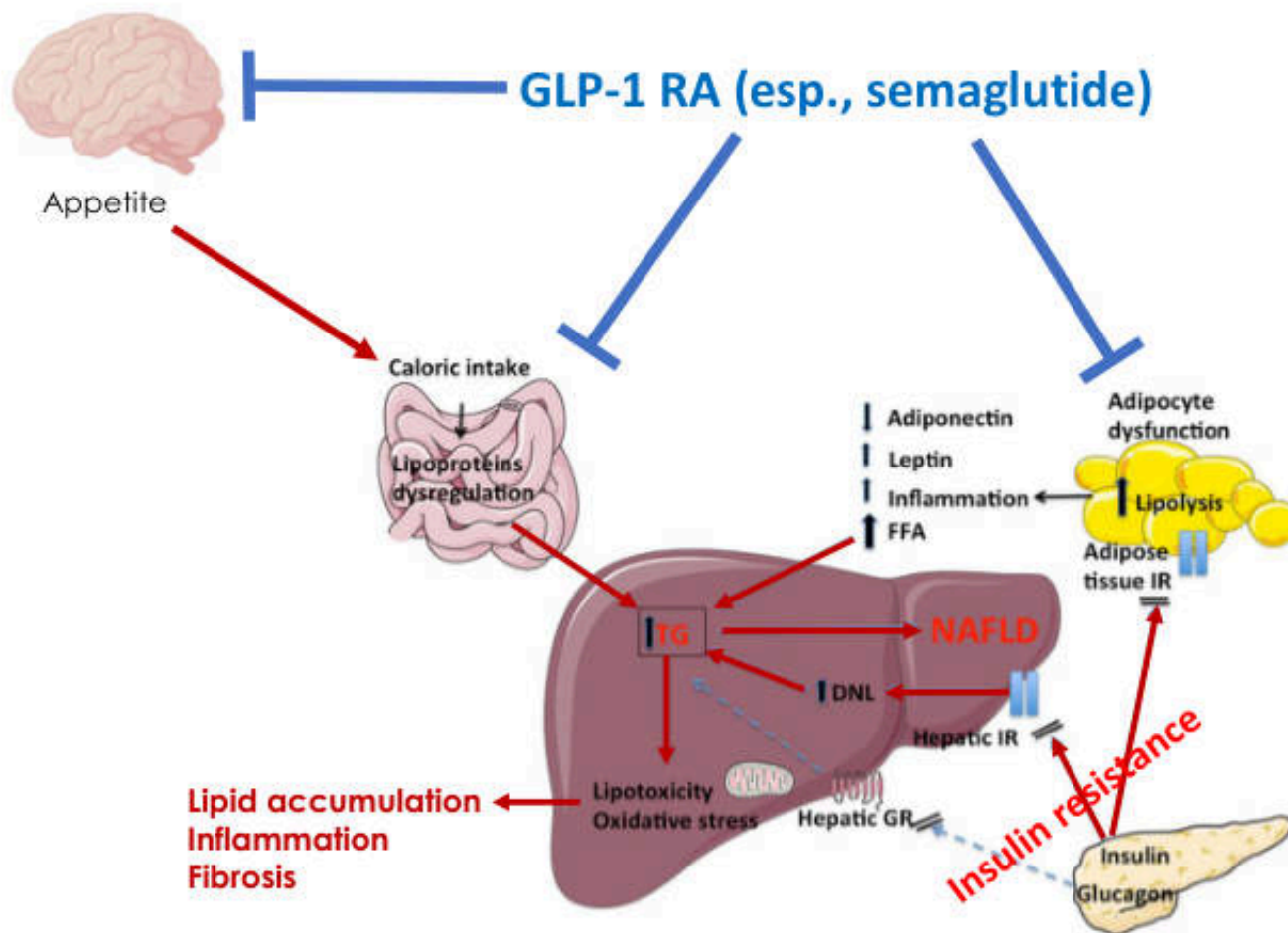


Clinical benefits of GLP-1R agonists and dual GIP/GLP-1R agonists

Type 2 diabetes		Weight loss in obesity	Metabolic dysfunction–associated steatohepatitis (MASH)	Obstructive sleep apnea in obesity
Cardiovascular	Renal			
↓ Major cardiovascular adverse event (MACE), e.g., myocardial infarction (MI) and ischemic stroke	↓ Declined renal function	Liraglutide (high dose) Semaglutide inj. and oral (high dose) Tirzepatide	Semaglutide inj.	Tirzepatide
Liraglutide Dulaglutide Semaglutide inj.	Liraglutide Dulaglutide Semaglutide inj.			

¹Cardiovasc Diabetol 2021;20:189. doi: 10.1186/s12933-021-01366-8, ²Front. Endocrinol. 14:1085799. doi: 10.3389/fendo.2023.1085799, ³Expert Opin Pharmacother. 2025;26(1):31-49, ⁴N Engl J Med. 2024;391(13):1193-1205, ⁵Diabetes Care 2026;49(Suppl. 1):S183–S215. doi: 10.2337/dc26-S009.

Effects of GLP-1 RA in MASH

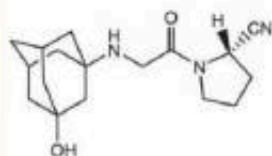


FFA= free fatty acids; TG= triglycerides;
IR= insulin resistance; GR= glucagon resistance
DNL= De novo lipogenesis

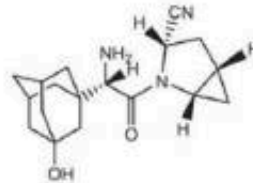
DPP-4 inhibitors

- Lower activity of DPP-4 by > 95% for 12 h
 - > 2-fold elevation of plasma concentrations of active GLP-1

Class 1

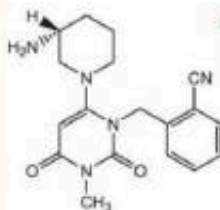


Vildagliptin

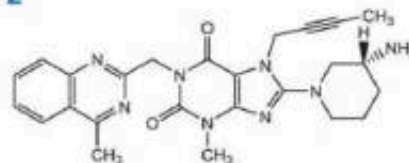


Saxagliptin

Class 2

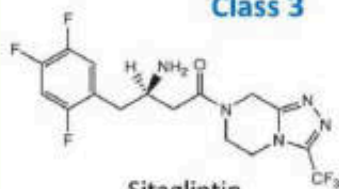


Alogliptin

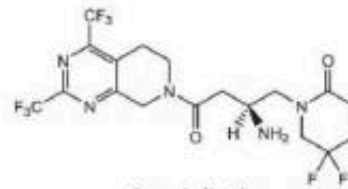


Linagliptin

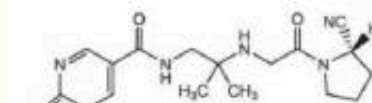
Class 3



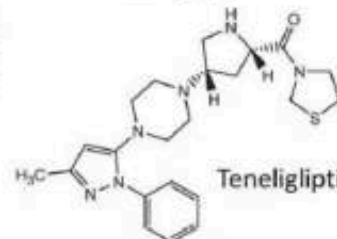
Sitagliptin



Gemigliptin



Anagliptin



Teneligliptin



Pharmacological properties of DPP-4 inhibitors

Parameters	Sitagliptin	Vildagliptin	Saxagliptin	Alogliptin	Linagliptin
Pharmacodynamics					
In vitro DPP-4 inhibition (IC ₅₀ in nM)	19	62	50	24	1
DPP-4 selectivity (vs DPP-8 and DPP-9)	>2,600	<100	<100	>14,000	>10,000
Pharmacokinetics					
Oral bioavailability (%)	87	85	75	70	30
Volume distribution (L)	198	71	151	300	368–918
Fraction bound to proteins (%)	38	9.3	<10	20	70
Half life (T _{1/2}) (h)	8–14	2–3 <i>twice daily</i>	2.2–3.8	12.4–21.4	120–184
Liver excretion (%)	13	4.5	22	13	85
Kidney excretion (%)	87	85	75	76	5
Proportion excreted unchanged (%)	79	23	24	95	~90
Substrate for CYP3A4/5	Low	No	Yes	No	No
Drug–drug interactions	Unknown	Unknown	CYP3A4/5 inducers and inhibitors	Unknown	Unknown

Strong CYP3A4 inhibitors

e.g., ketoconazole, atazanavir, clarithromycin, indinavir, itraconazole, nefazodone, nelfinavir, ritonavir, saquinavir, and telithromycin



ADRs of incretin-based therapy

Side effect	Frequency		Potential mechanism
	GLP-1RA	DPP-4I	
& Tirzepatide			
GI			
Nausea	+++	++	Gastric emptying ↓, activation nausea centres
Vomiting	++	+	Gastric emptying ↓, activation nausea centres
Diarrhoea	+++	+	Unknown
Constipation	++	+	Intestinal motility ↓
Flatulence	++	+	Unknown
Gastric reflux	++	+ / ++	Unknown
Pancreatitis	+ / -	+ / -	Unknown
Cholelithiasis	+ / -	?	Weight loss, bile acid production ↓, gallbladder motility ↓
Other			
Hypoglycaemia*	+ / ++	+ / ++	Insulin secretion ↑, intestinal glucose uptake ↓
Nasopharyngitis	+	+	Unknown
Anaphylaxis	+ / -	+ / -	Immunoreactive
Prerenal failure	+ / -	+ / -	Dehydration by vomiting, diuresis ↑

Based on the summary product characteristics of all registered GLP-1RA and DPP-4I

***Predominantly observed in combination with sulfonylurea agents, not with metformin**

+++ , very common (>1/10)

++ , common (>1/100)

+ , uncommon (>1/1000)

+ / - , rare to very rare (>1/10 000)

- , not present

?, unknown

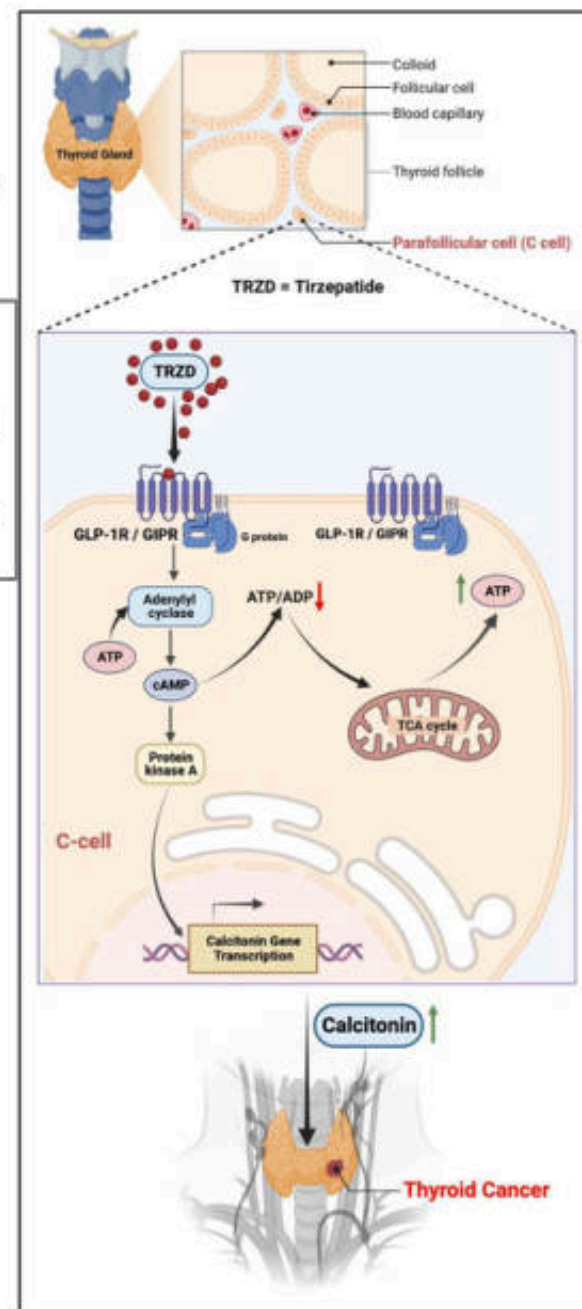
DPP-4I, dipeptidyl peptidase 4 inhibitor; GLP-1RA, glucagon-like peptide-1 receptor agonist

Exenatide, and probably lixisenatide should not be given to persons with moderate-to-severe renal failure

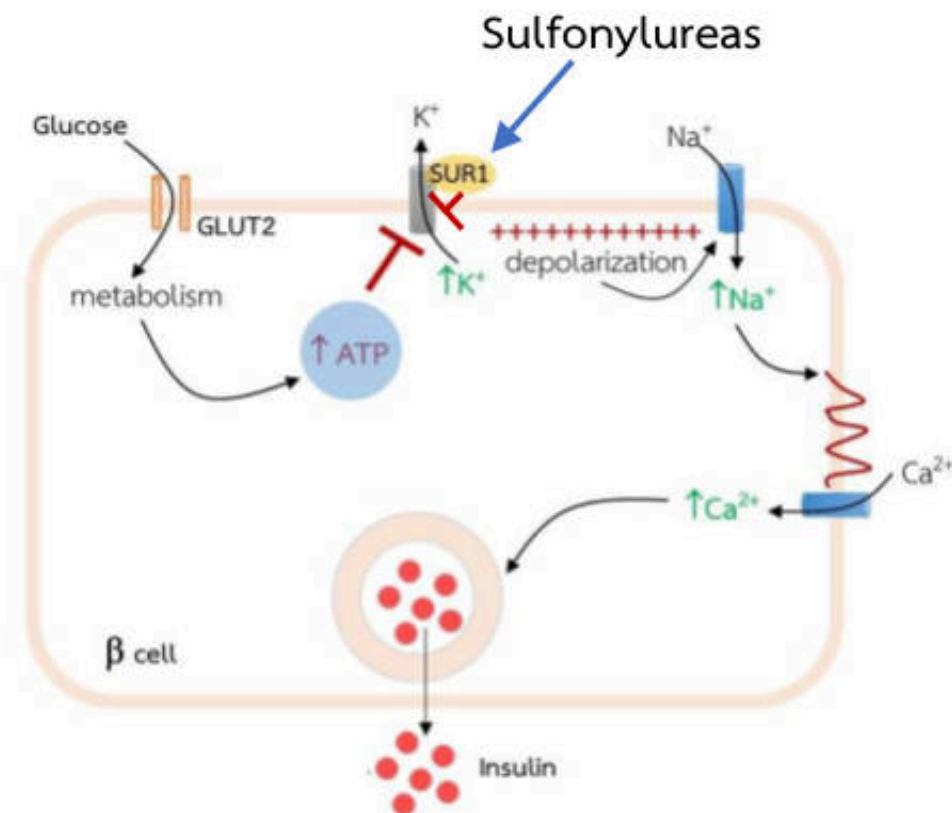
US FDA Warning

- GLP-1R agonists and tirzepatide are contraindicated in patients with a personal or family history of medullary thyroid carcinoma (MTC) or in patients with Multiple Endocrine Neoplasia syndrome type 2 (MEN 2)
 - Thyroid C-cell tumors in rats.
 - No known in humans

The sustained activation of GIP/GLP-1 receptors is accompanied by C-cell hyperplasia and transformation into C-cell adenomas and medullary thyroid carcinoma (MTC)



Sulfonylureas (SUs)



Adapted from Diabetes 2016;65:2479–2481

• First generation

- Tolbutamide
- Chlorpropamide
- Tolazamide

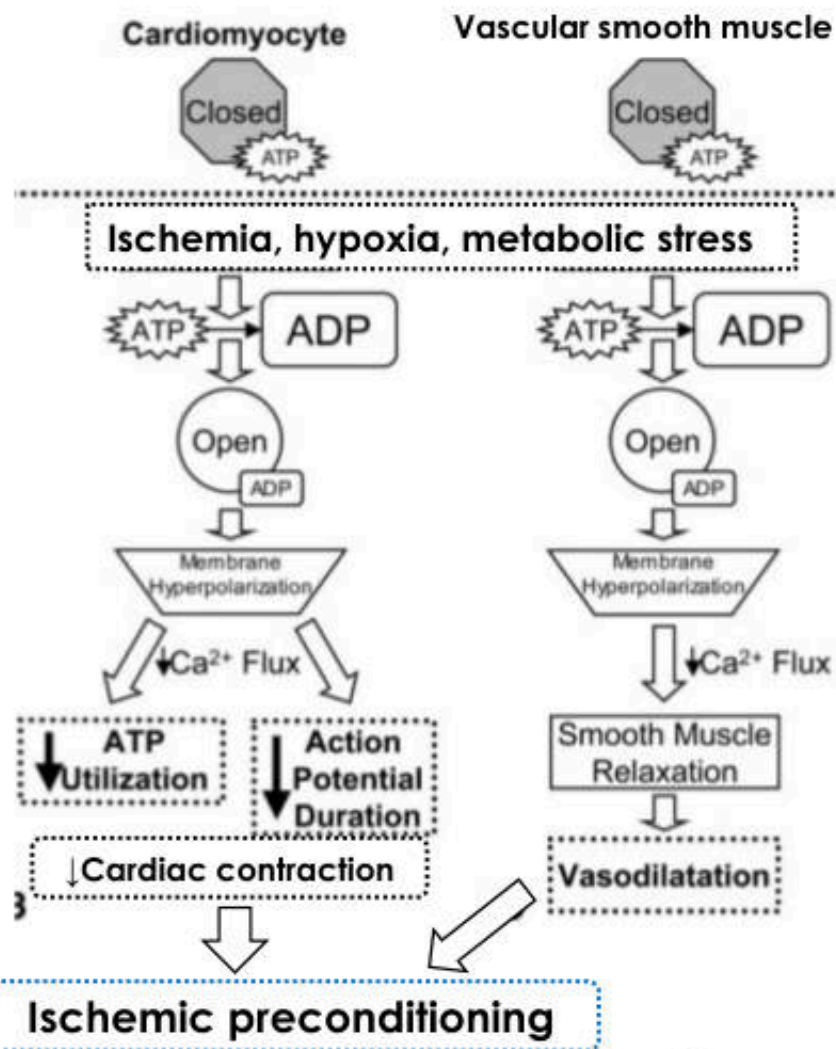
• Second generation (more potent)

- Glibenclamide (glyburide)
- Gliclazide
- Glipizide
- Glimepiride
- Glipizide
- Gliquidone



Physiological effects of SUR-mediated K_{ATP} channel function in selected cells

Isoform*	Tissue Specificity‡	Primary Function
SUR1	Brain; pancreas	Pancreatic β -cells: regulating insulin release
SUR2A	Heart	<u>Myocardial cells</u> : regulating action potential duration
SUR2B	Heart; eye; brain; cerebellum; colon	<u>Smooth muscle cells, including vascular</u> : regulating action potential duration and vasodilation





Selectivity of SUs and non-SUs on SUR subtypes

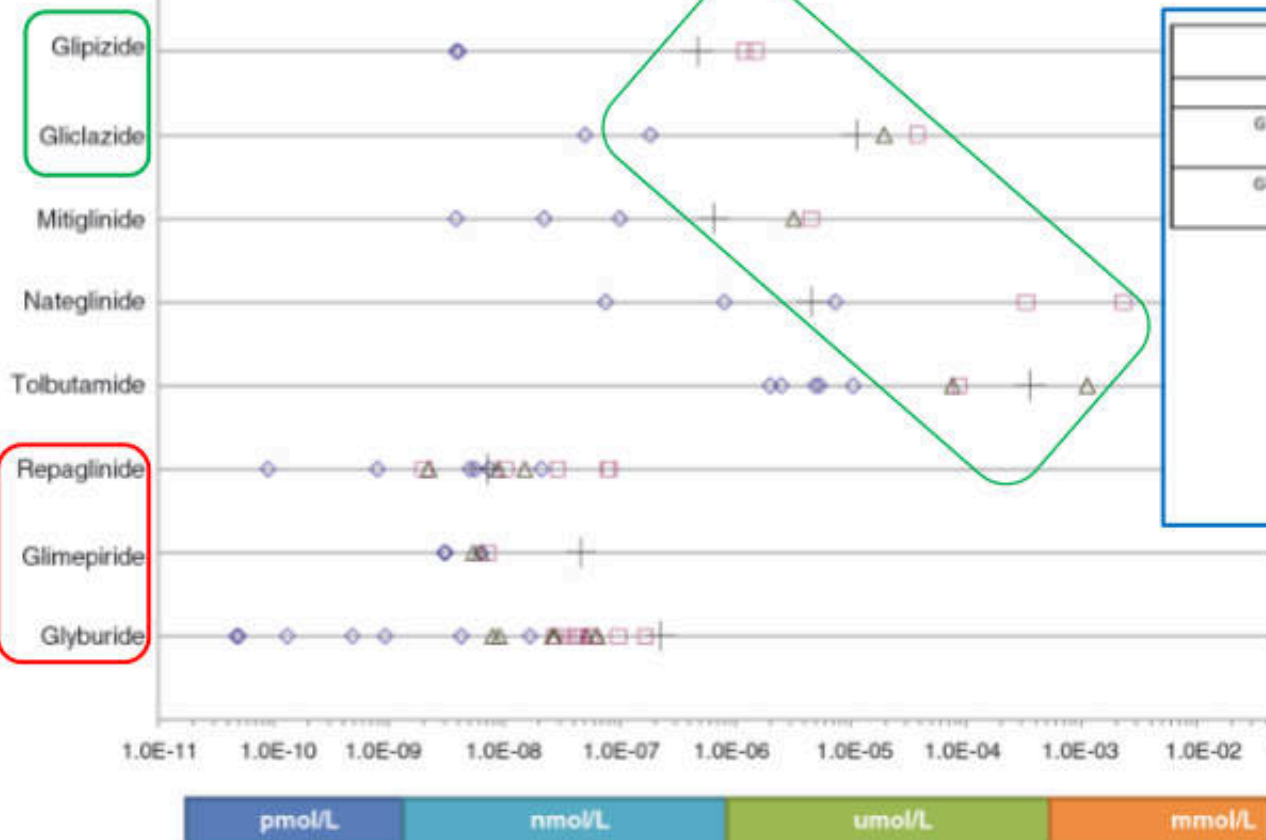
◇ SUR1 (Pancreatic Receptor)

△ SUR2A (Cardiac Myocyte Receptor)

□ SUR2B (Smooth Muscle Receptor)

+ Steady State Concentration

Hazard ratios (HR) for [major adverse cardiovascular events](#) among new users of gliclazide and glyburide



	MACE n(per 1000 person years)		Hazard Ratio HR(95% Confidence Interval)
	Low Dose	High Dose	
Gliclazide	302 (29.2)	336 (32.2)	1.10 (0.94 - 1.29)
	206 (28.2)	288 (32.8)	1.15 (0.96 - 1.38)
Glyburide	278 (32.8)	326 (38.6)	1.18 (1.00 - 1.38)
	185 (31.5)	247 (38.9)	1.24 (1.02 - 1.50)

Diabetes Obes Metab. 2019;21:761-771.

Comparative pharmacological properties of sulfonylureas

PK/PD properties	Glibenclamide	Gliclazide	Glipizide	Glimepiride	Glipizide XL	Gliclazide MR
Duration of action	16-24 h	10-24 h	12-24 h	24 h	>24 h	>24 h
Volume of distribution	9-10 L	13-24 L	10-11 L	19.8-37.1 L	10 L	19 L
Protein binding (%)	99	85-99	98-99	99	98-99	>90
Metabolism	Hepatic	Hepatic (no active metabolites)	Hepatic (no active metabolites)	Hepatic (active metabolites)	Hepatic (no active metabolites)	Hepatic, (no active metabolites)
Bioavailability (%)	99	80	100	100	100	97
Half-life	10 h	8-12 h	2-5 h	5±4.1 h	2-5 h	16 h
Time to peak	2-4 h	2-4 h	1-3 h	2-3 h	6-12 h	6-7 h
Excretion	50% renal	80% renal	80% renal	60% renal	80% renal, 10% feces	<60-70% renal 10-20% in feces
Drug-drug interaction	May interact with CYP2C9 inducers or inhibitors					
PK changes in elderly	Slow elimination; higher volume of distribution	Likely increase half-life and slower elimination	No significant differences in PK properties			
PK changes in renal and hepatic impairment	May be altered increasing the risk of toxic reactions to drug	May affect the distribution and may also reduce the capacity for neoglucogenesis	Metabolism and excretion may be slowed	No significant differences in PK properties in renal impairment while it is not evaluated in hepatic impairment	May affect the disposition of drug and also diminish gluconeogenic capacity	May affect the distribution and may also reduce the capacity for neoglucogenesis increasing the risk of hypoglycemia

MR: Modified release, XL: Extended release, PK/PD: Pharmacodynamic/pharmacokinetic, SUs: Sulfonylureas.



ADRs and drug interactions of SUs

- **ADRs**

- [Hypoglycemic reactions, including coma](#)
- [Weight gain of 1–3 kg is common](#)
- Less frequent side effects: nausea and vomiting, cholestatic jaundice, agranulocytosis, aplastic and hemolytic anemias, hypersensitivity reactions
- Rarely: an alcohol-induced flush (esp. chlorpropamide)

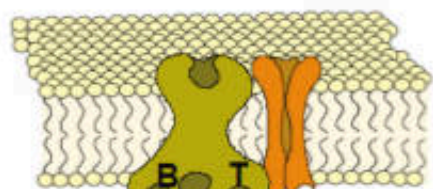
- **Drug interactions**

- Ethanol may enhance the action of SUs and cause hypoglycemia
- Some drugs (sulfonamides, clofibrate, and salicylates) displace the SUs from binding proteins → transiently increase the concentration of free drug

Drug	Mechanism	Effect
Fluconazole, ketoconazole, voriconazole, amiodarone, co-trimoxazole (sulfamethoxazole component), fluoxetine, fluvoxamine, metronidazole	Inhibition of CYP2C9	Hypoglycemia
Carbamazepine, phenobarbital, rifampin, St. John's wort, dexamethasone	Induction of CYP2C9	Reduced efficacy (hyperglycemia)

Non-sulfonylureas

SUR1/ Kir6.2



Repaglinide

Glibenclamide
Glimepiride(?)

Tolbutamide
Gliclazide
Acetohexamide
Tolazamide
Glycopyramide
Chlorpropamide

Nateglinide
Mitiglinide

B = benzamide-binding site
T = tolbutamide-binding site

- Rapid onset (15-30 min)
 - Major therapeutic effect is reducing postprandial glycemic elevations
- Short $t_{1/2}$ (~1 h)
 - Multiple preprandial use
- **Nateglinide**
 - Metabolized by CYPs (2C9, 70%; 3A4, 30%)
 - Fewer episodes of hypoglycemia
- **Repaglinide**
 - Metabolized by CYP3A4 to inactive derivatives

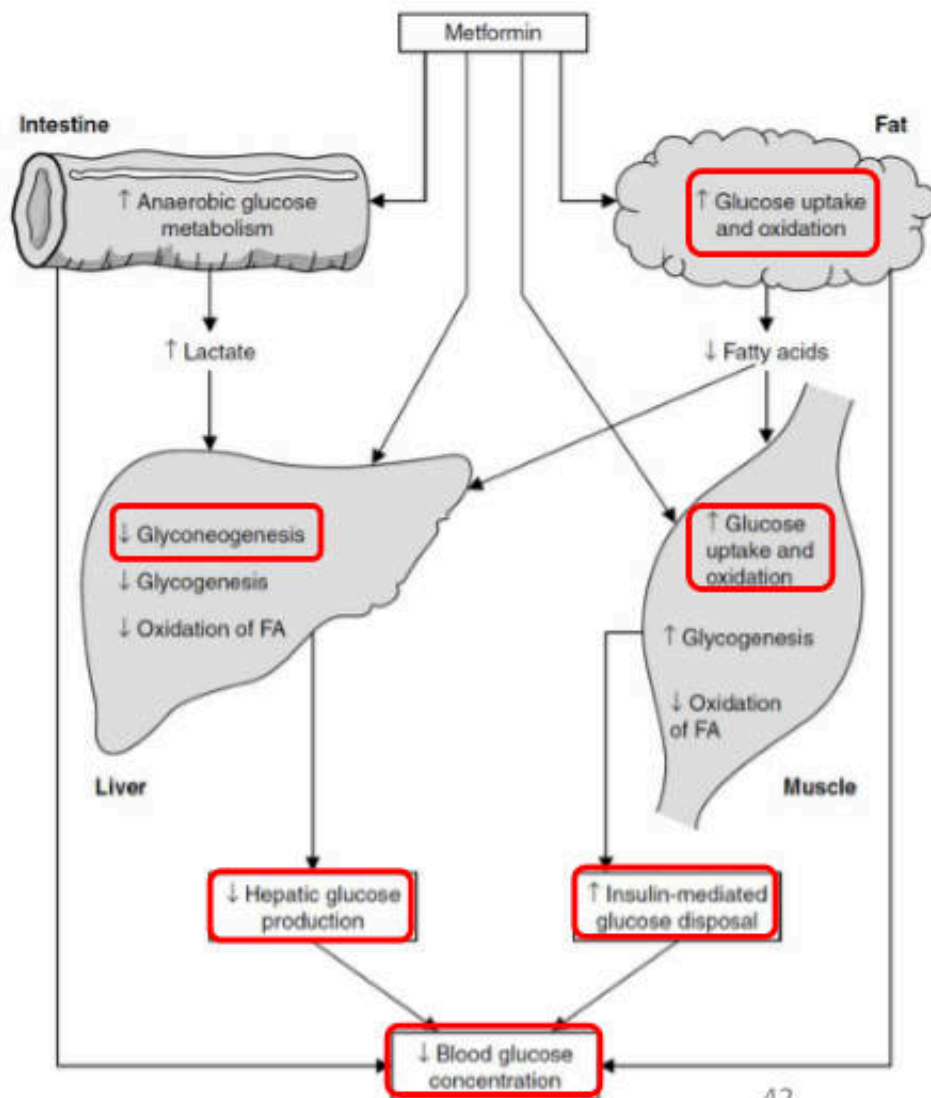
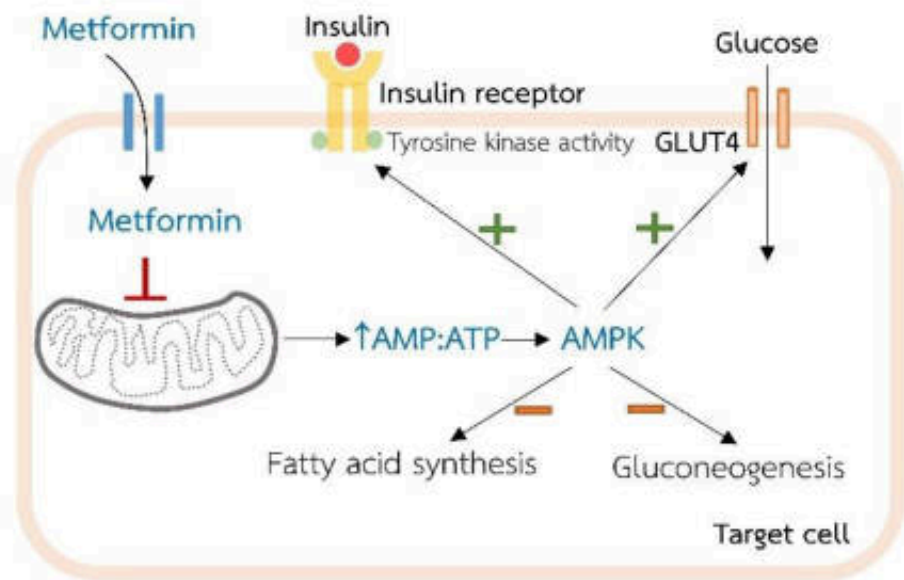


Insulin sensitizers

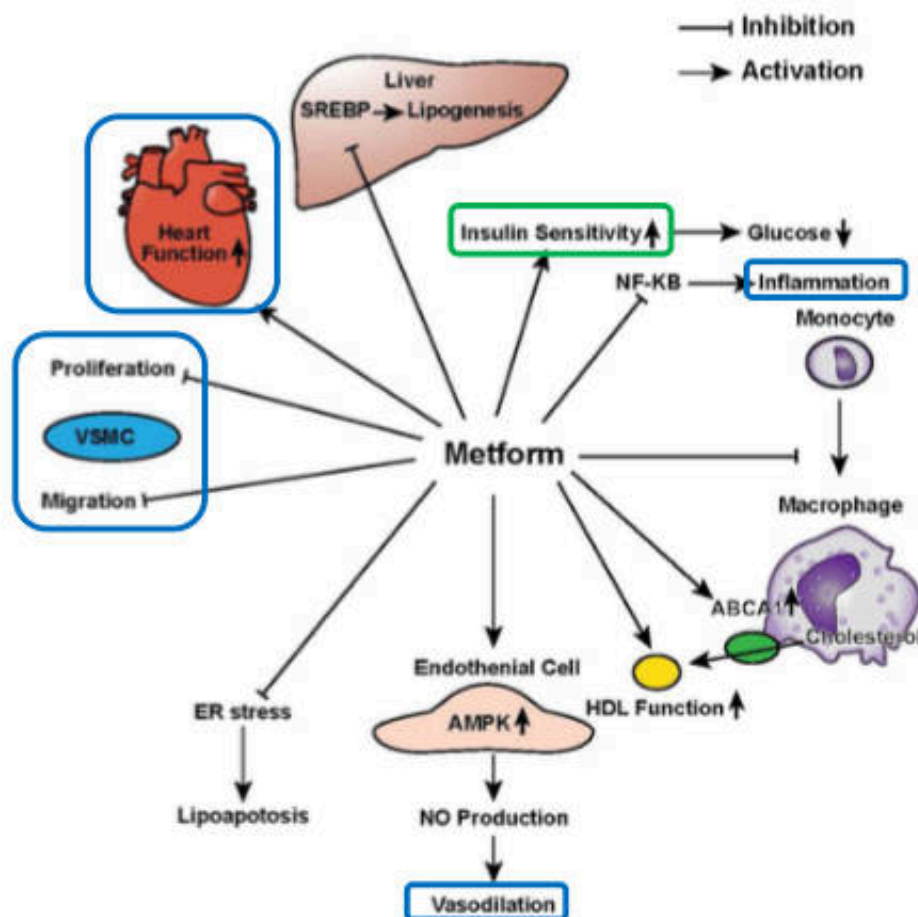
- Biguanides
- Thiazolidinediones (TZDs)

Metformin

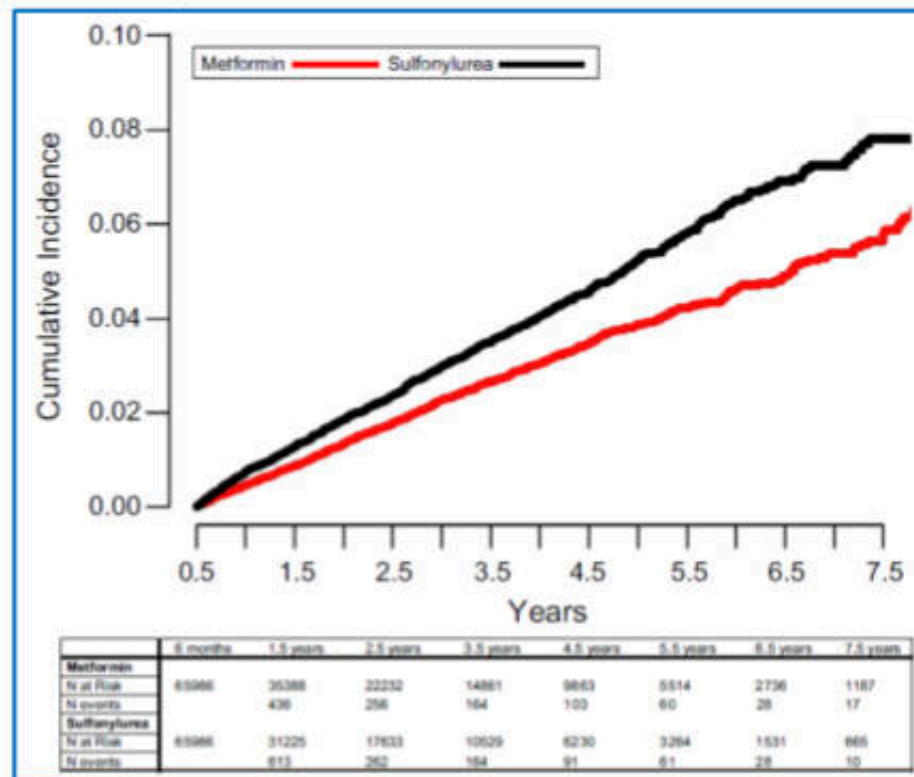
- The only member of the biguanide class available for use today
- **The first-line drug for type 2 diabetes**
- Used off-label to treat polycystic ovarian syndrome



Pleiotropic (cardioprotective) effects of metformin



Cumulative incidence of heart failure hospitalization or cardiovascular death over time



J Am Heart Assoc. 2017;6:e005379.

Atherogenic process



Pharmacokinetics of metformin

- Absorbed primarily from the small intestine
 - Bioavailability: 70–80%
 - T_{max} after an oral dose occur at about 2 h
 - Plasma $t_{1/2}$ is 4–5 h
- Does not bind to plasma proteins
- **Excreted unchanged in the urine**
 - Dose adjustment in renal impairment
 - Reduces risk of lactic acidosis



ADRs and drug interactions of metformin

- **ADRs**

- **Most common (10-25%)**

- GI: nausea, indigestion, abdominal cramps or bloating, diarrhea
 - Interference with the absorption of glucose and bile salts
 - 20%–30% lower blood levels of vitamin B12
 - Neutral activity for body weight (some evidence shows weight loss)
 - Rarely causes hypoglycemia
 - Lactic acidosis (rare but severe)
 - **Metformin inhibits mitochondrial function → overproduction of lactate**
 - Reported in patients with the setting of concurrent conditions that can cause poor tissue perfusion (e.g., renal failure, sepsis, myocardial infarction, and congestive heart failure)

- **DIs**

- Cationic medications: cimetidine, furosemide, and nifedipine compete renal tubular transport systems
 - Reduce metformin excretion



Thiazolidinediones (TZDs)

- **Pioglitazone** and rosiglitazone
 - Increase insulin-mediated glucose uptake by 30%–50% in patients with type 2 diabetes
- The onset of action is relatively slow
 - Maximal effects on glucose homeostasis develop gradually over the course of 1–3 months

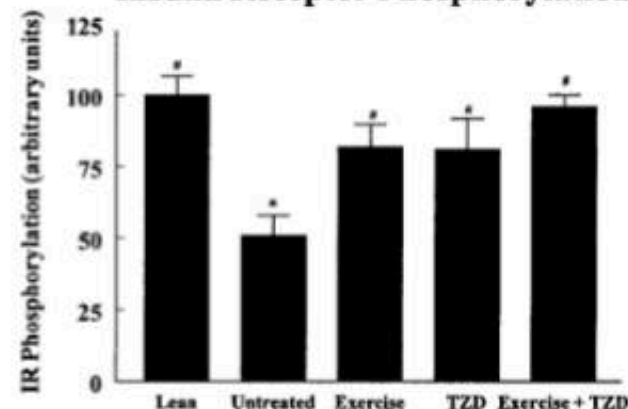
Adipose tissue	Muscle	Liver
↑ Glucose uptake	↑ Glucose uptake	↓ Gluconeogenesis
↑ Fatty acid uptake	↑ Glycolysis	↓ Glycogenolysis
↑ Lipogenesis	↑ Glucose oxidation	↑ Lipogenesis
↑ Pre-adipocyte differentiation	↑ Glycogenesis ^a	↑ Glucose uptake ^a
a Inconsistent findings.		
↑ indicates increase; ↓ indicates decrease.		



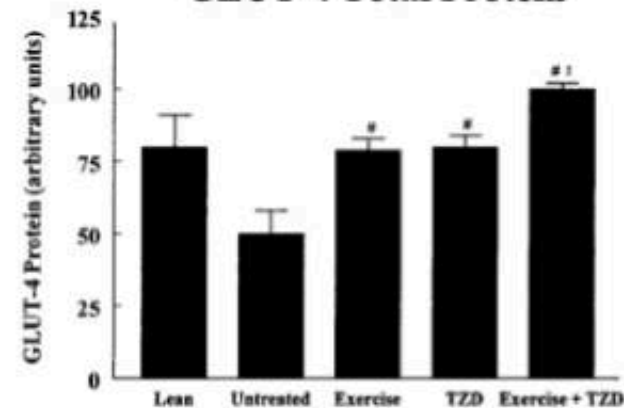
Thiazolidinediones (TZDs)

Muscle of Zucker fatty rat

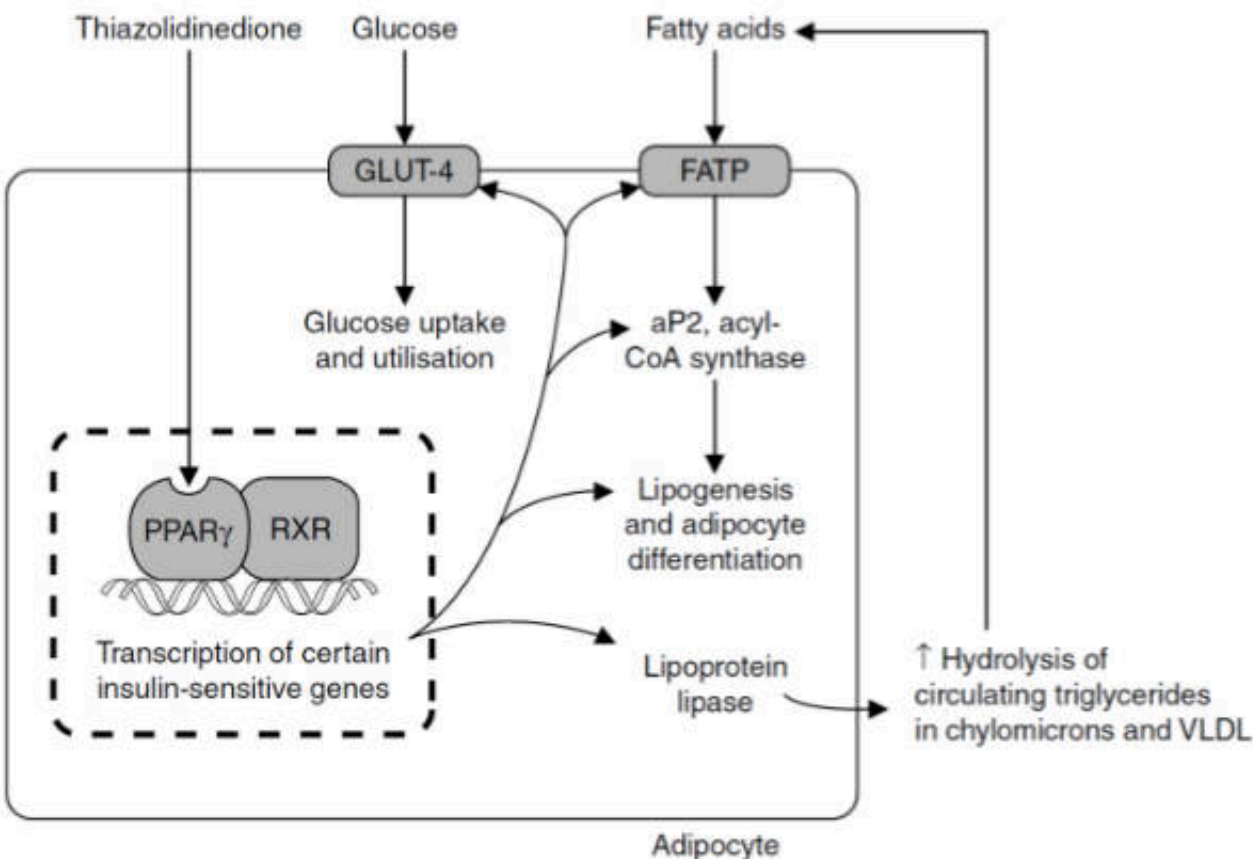
Insulin Receptor Phosphorylation



GLUT-4 Total Protein



Diabetes 2000;49:2154-2159





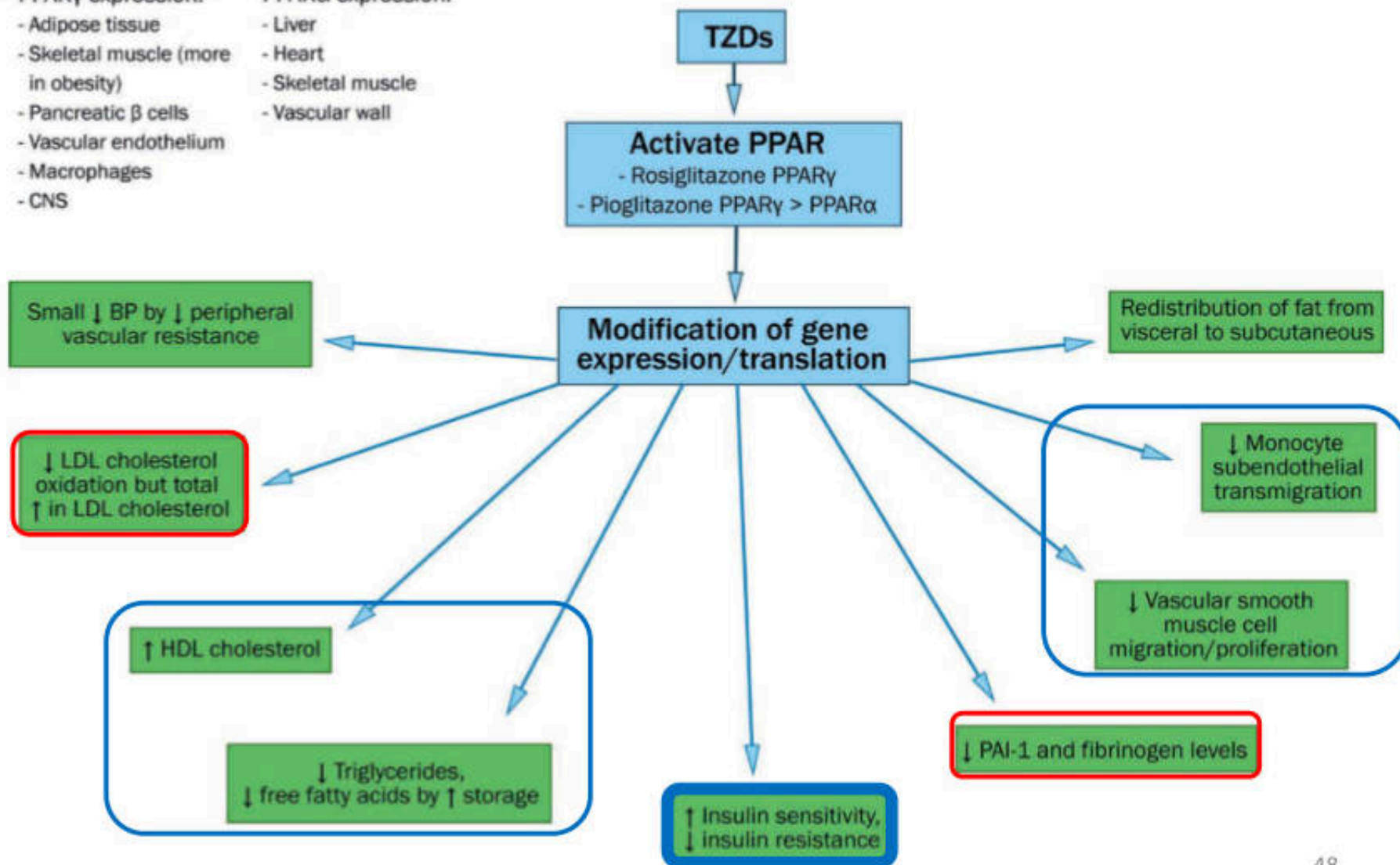
Thiazolidinediones (TZDs)

PPAR γ expression:

- Adipose tissue
- Skeletal muscle (more in obesity)
- Pancreatic β cells
- Vascular endothelium
- Macrophages
- CNS

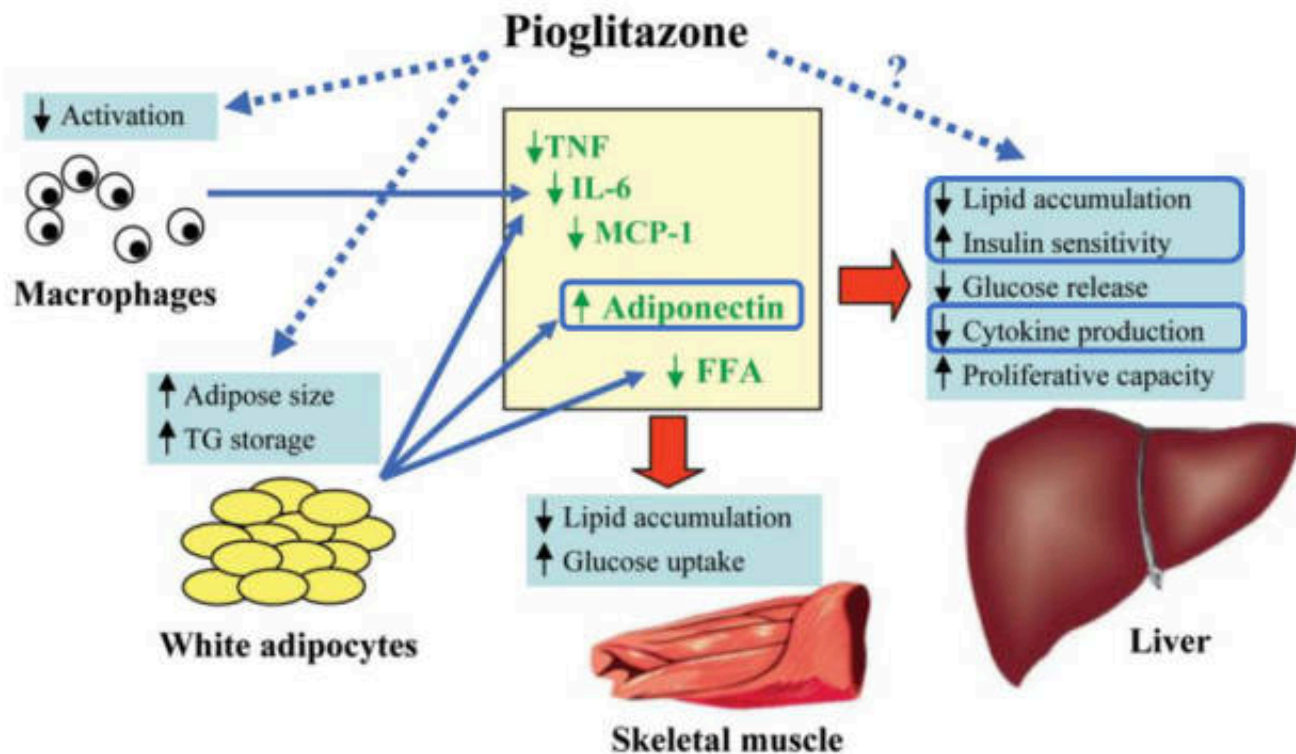
PPAR α expression:

- Liver
- Heart
- Skeletal muscle
- Vascular wall



Pioglitazone in MASH

- It improves liver histology
 - Reduces steatosis, inflammation, and **fibrosis?**
- By increasing adiponectin, enhancing insulin sensitivity, and reducing fatty acid influx into the liver



Hepatology. 2009;49(5):1427–1430. doi: 10.1002/hep.22983



Pharmacokinetics of TZDs

- Metabolism
 - Pioglitazone: CYP2C8 and CYP3A4
 - Rosiglitazone: CYP2C8/9
- Half life: Pioglitazone (3 to 7 hrs), rosiglitazone (3 to 4 hrs)
- Dose adjustment is not required for elderly patients or patients with renal insufficiency
- Withheld in patients with liver impairment



ADRs and drug interactions of TZDs

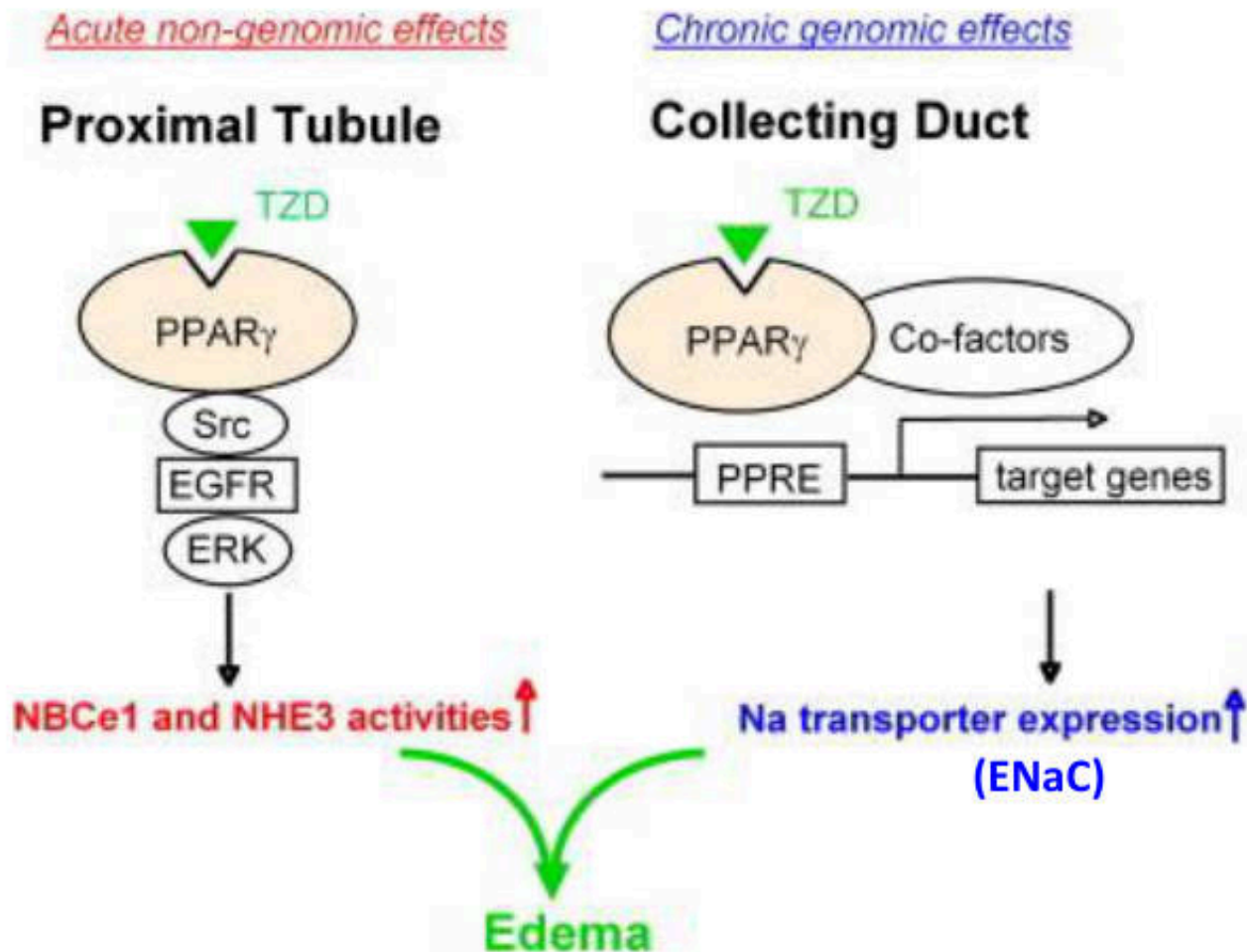
- **Weight gain (2-4 kg)**

	<i>Anemia (%)</i>	<i>Edema (%)</i>	<i>LDL cholesterol increase (%)</i>
Rosiglitazone	+2	+4.8	+18
Pioglitazone	+1	+4.8–15.3	+19

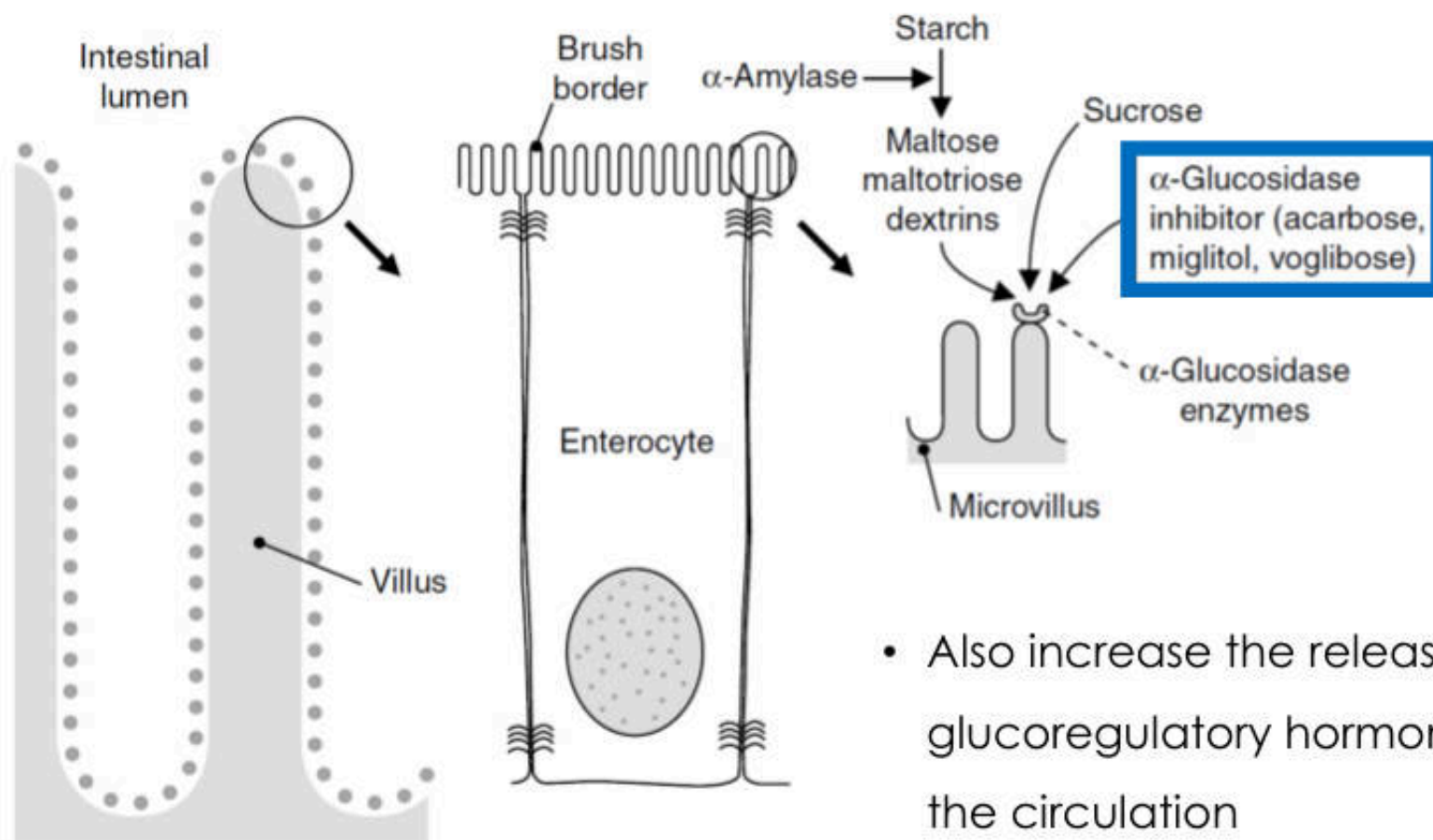
Diabetes Technol Ther. 2000;2(3):429-40.

- **Exposure to TZDs over several years → ↑ risk of heart failure (2-fold)**
 - Should not be used in patients with moderate-to-severe heart failure
- In the past, evidence suggested that rosiglitazone increased the risk of cardiovascular events (myocardial infarction, stroke)
 - But this effect has now been less concerned
- Increased risk of bone fracture in women
- **There are no known drug–drug interactions**

Possible mechanism of TZDs-induced edema



α -glucosidase inhibitors



- Also increase the release of the glucoregulatory hormone GLP-1 into the circulation
- May contribute to their glucose-lowering effects

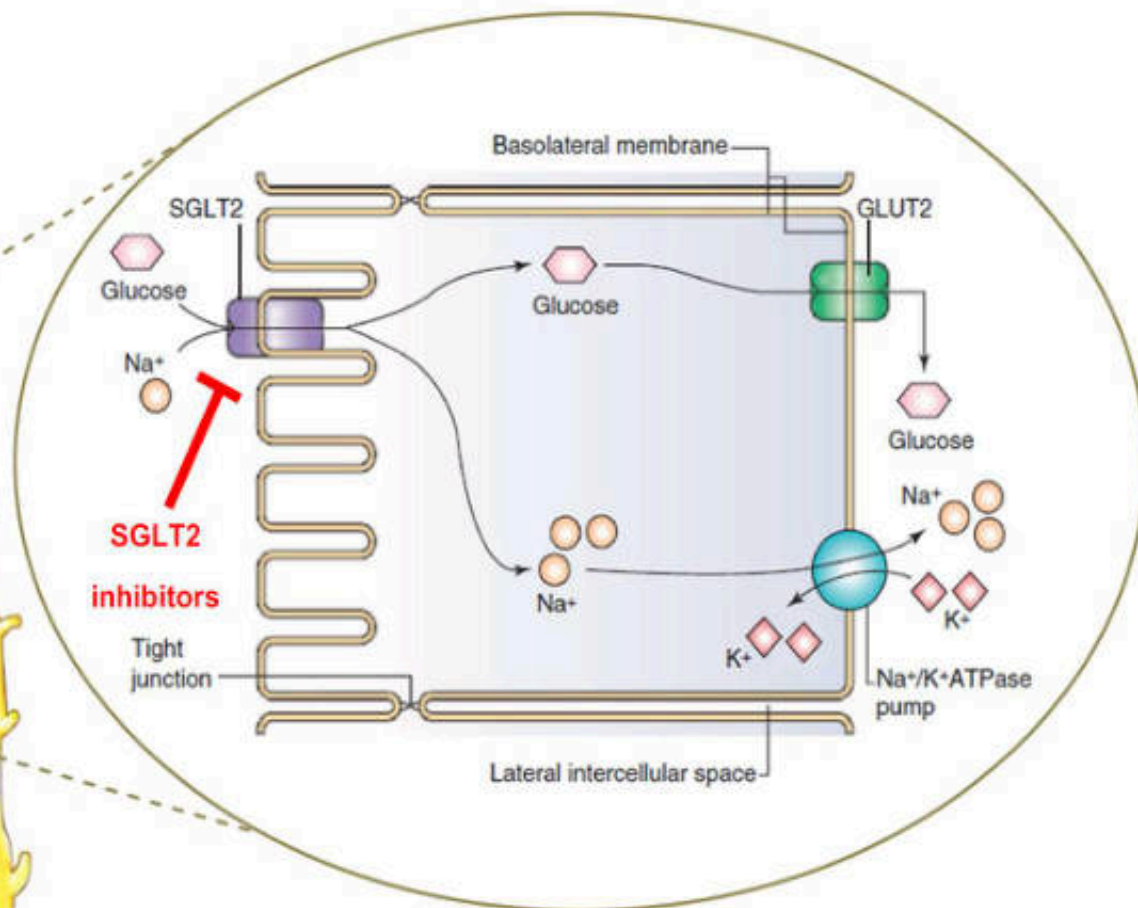
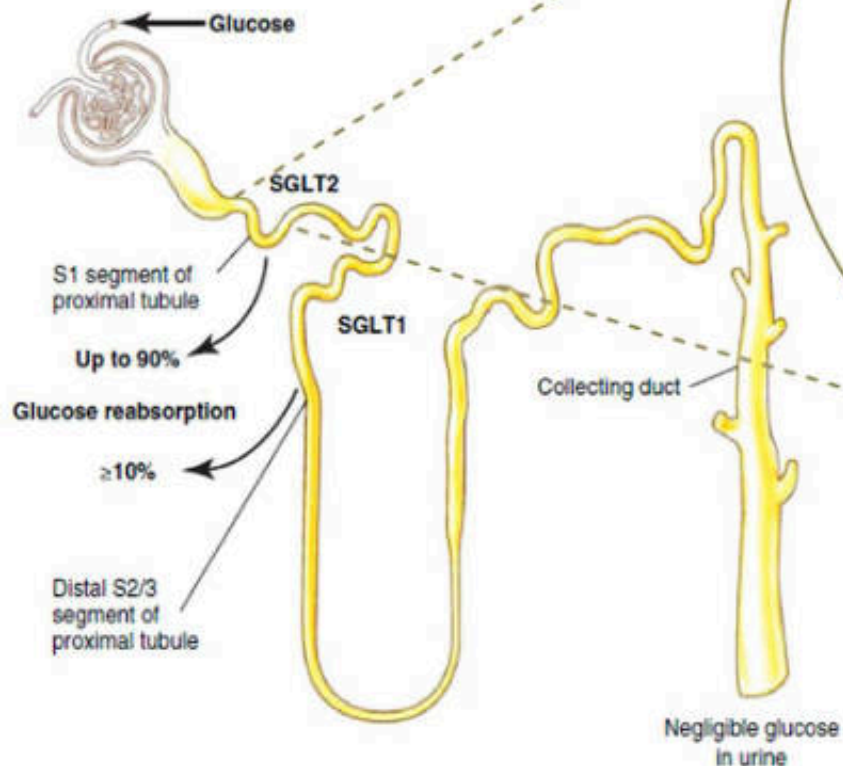


α -glucosidase inhibitors

- Taken before meals
- Acarbose is minimally absorbed; cleared by the kidney
- Miglitol absorption is saturable, with 50%–100% of any dose entering the circulation; cleared by the kidney
 - Dose reductions for patients with severe renal impairment
- **ADRs**
 - Most prominent adverse effects: malabsorption, flatulence, diarrhea, and abdominal bloating
 - Hypoglycemia has been described when added to insulin or an insulin secretagogue
- **DIs**
 - Acarbose can decrease the absorption of digoxin
 - Miglitol can decrease the absorption of propranolol and ranitidine

SGLT2 inhibitors

160-180 g of
glucose is
filtered daily



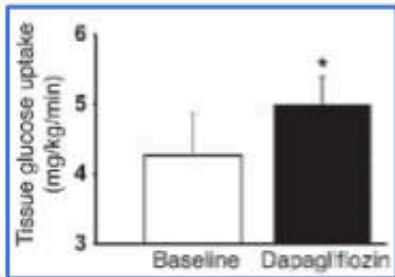


Selectivity and potency of SGLT2 inhibitors

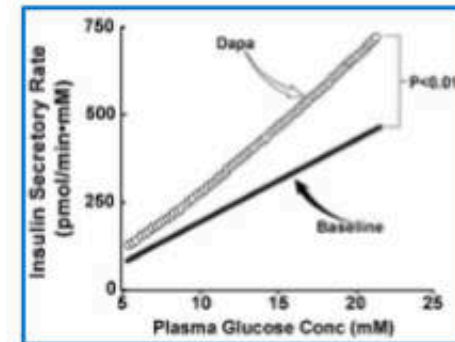
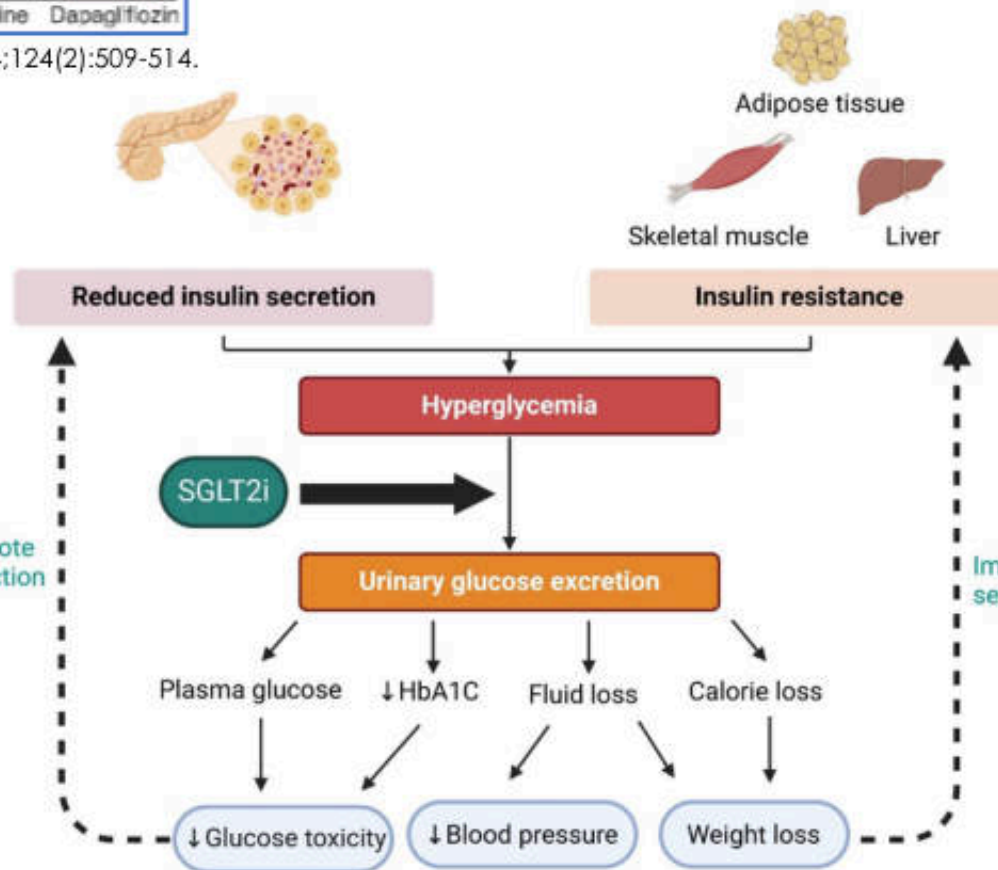
Drugs	In vitro study		Human dose
	SGLT2 selectivity (IC ₅₀ for SGLT1:SGLT2)	Potency of SGLT2 inhibition (IC ₅₀ , nM)	
Canagliflozin	160	2.7	100-300 mg OD
Dapagliflozin	1400	1.2	5-10 mg OD
Empagliflozin	2700	3.1	10-25 mg OD
Ertugliflozin	2200	0.9	5-15 mg OD
Bexagliflozin	2435	2.0	20 mg OD
Sotagliflozin	20	1.8	200-400 mg OD



Summary of pharmacological effects



J Clin Invest. 2014;124(2):509-514.



J Clin Endocrinol Metab. 2015;100:1927-32.



Summary of pharmacological effects of SGLT2 inhibitors (Cont.)

- Glycemic control

- Urinary glucose excretion: 50-90 g/day
 - 30-50% of total filtered glucose
- HbA_{1c} reduction (placebo-subtracted): 0.5-1.0%
- FPG reduction: 20-40 mg/dl

- Blood pressure reduction

- ↓ SBP : 2-5 mmHg
- ↓ DBP : 1-2 mmHg

- Weight loss: 1-3 kg (plateau after 6 months)



1. Hemodynamic effect



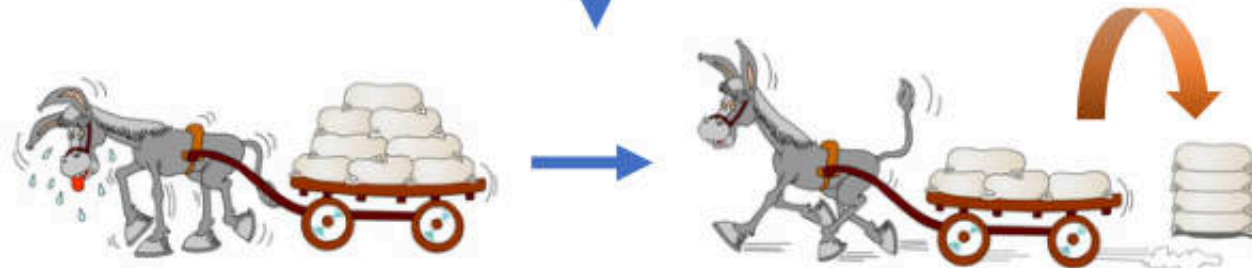
Diuretic and natriuretic effects $\dashrightarrow$ Blood pressure lowering



Reduce preload



Reduce afterload



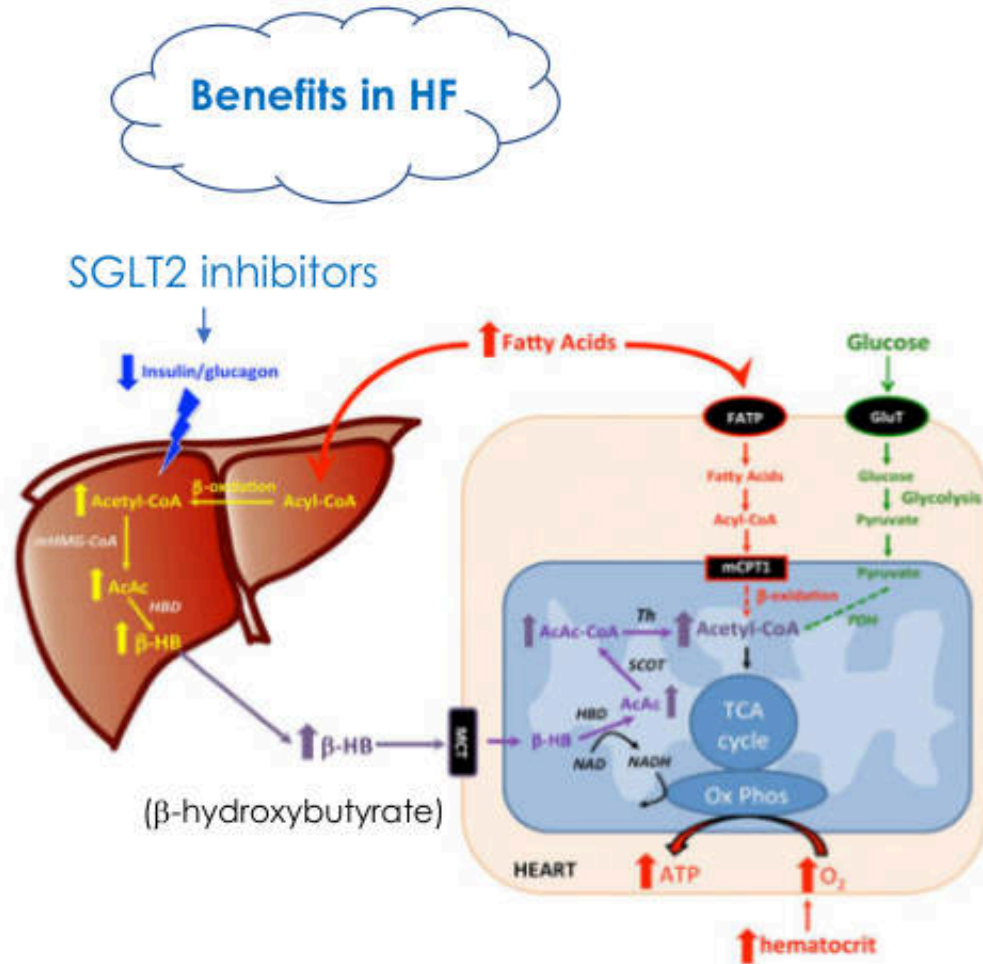
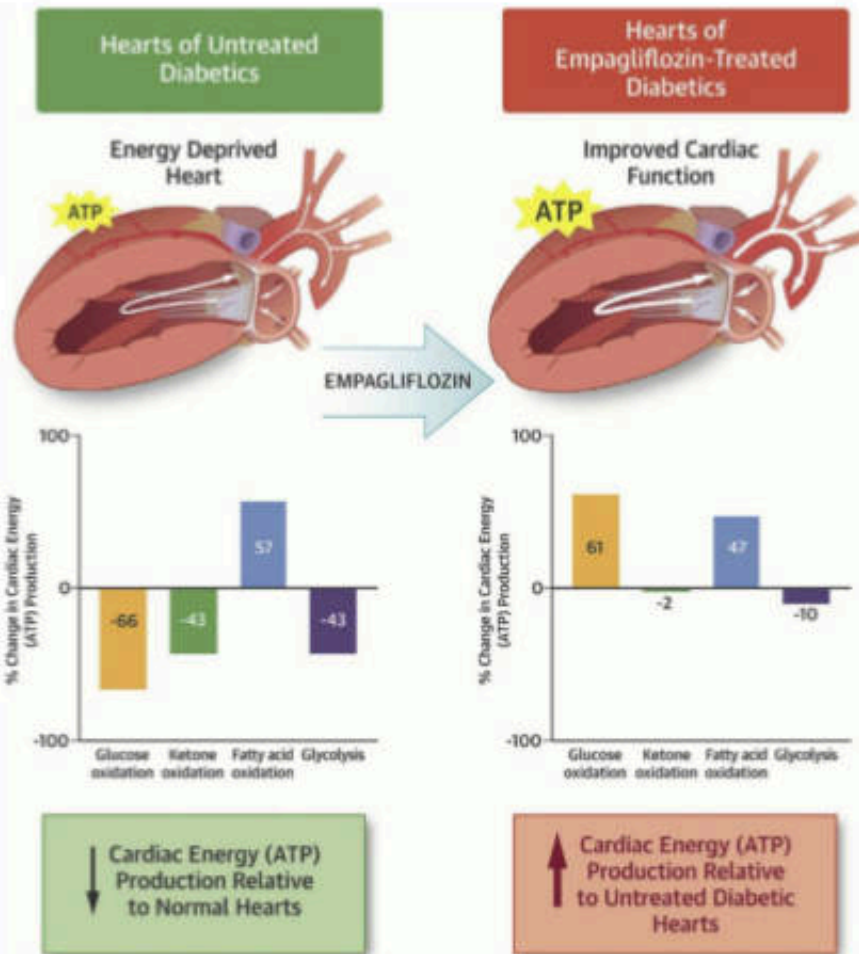
↓ Energy demand

Improve cardiac function



Possible mechanisms of cardiac benefits mediated by SGLT2 inhibitors (Cont.)

2. Cardiac energy production



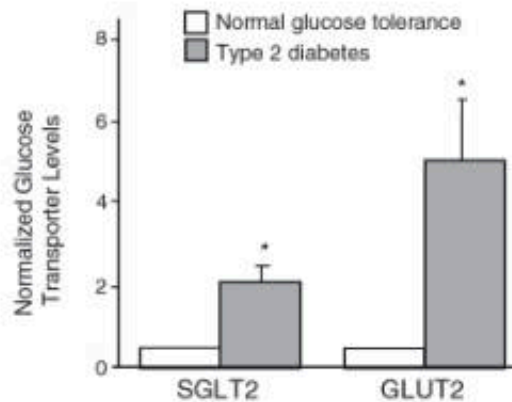
Diabetes Care. 2016;39:1108–14.



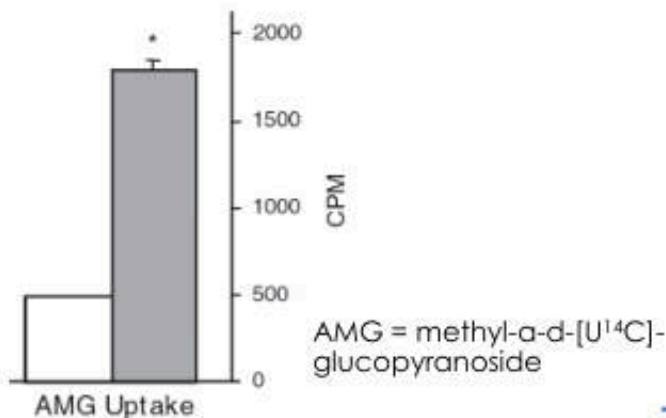
Possible mechanisms of renal benefits mediated by SGLT2 inhibitors

- In cultured renal proximal tubular epithelial cells

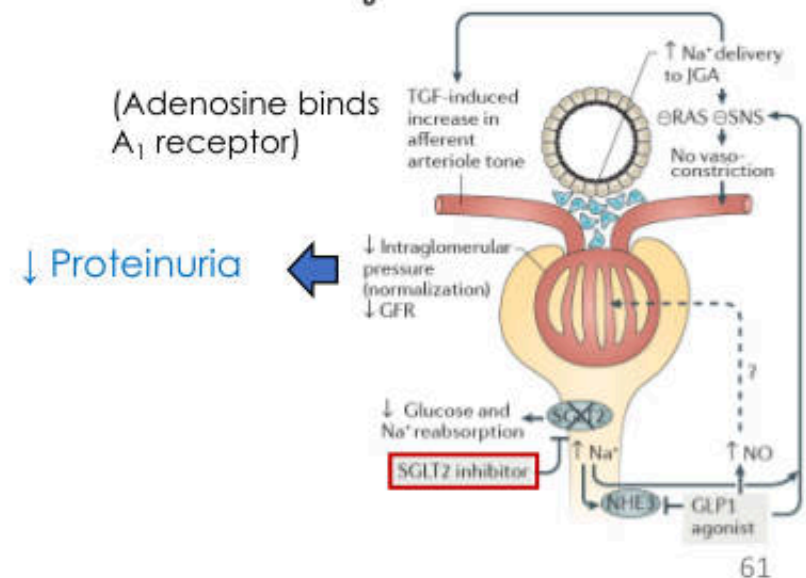
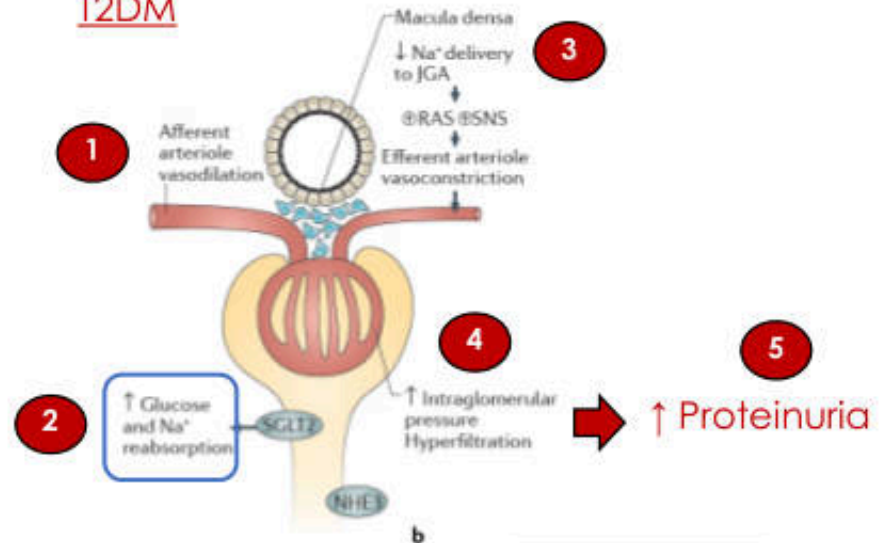
SGLT2 and GLUT2 Levels



Renal glucose uptake



T2DM





Clinical benefits of SGLT2 inhibitors

Type 2 diabetes

Cardiovascular

↓ Major cardiovascular adverse event (MACE), e.g., myocardial infarction (MI) and ischemic stroke

Empagliflozin
Canagliflozin
Dapagliflozin (only in patients with prior MI)

↓ Cardiovascular death and hospitalization for heart failure

All SGLT2i, except bexagliflozin (no study)

Renal

↓ Declined renal function

All SGLT2i, except ertugliflozin (no benefits) and bexagliflozin (no study)

Heart failure

↓ Cardiovascular death and hospitalization for heart failure

- Heart failure with preserved ejection fraction (HFpEF)

- Heart failure with reduced ejection fraction (HFrEF)

Empagliflozin

Dapagliflozin

Sotagliflozin

Chronic kidney disease

↓ Disease progression and death due to cardiovascular and/or renal causes

Empagliflozin
Dapagliflozin

↓ Cardiovascular death and hospitalization for heart failure

Sotagliflozin



Pharmacokinetics of SGLT2 inhibitors

	Canagliflozin	Dapagliflozin	Empagliflozin	Ertugliflozin	Bexagliflozin	Sotagliflozin
Oral BA (%)	65	78	>60	100	78	25
Protein binding (%)	98	91	86	95	93	>93
Vd (L)	119	118	74	85	262	9000
Metabolism	UGT1A9, 2B4	Mainly by UGT1A9	UGT2B7, 1A3, 1A8, 1A9	UGT1A9, 2B7	UGT1A9	UGT1A9
Plasma $t_{1/2}$ (h)	10-13	12.9	12.4	11-17	12	13.5-20.7
Excretion	Kidney: ~30% (mainly inactive) Feces: ~50% (mainly unchanged)	Kidney: ~70% (mainly inactive) Feces: ~20% (mainly unchanged)	Kidney: ~50% (half is unchanged) Feces: ~40% (mainly unchanged)	Kidney: ~50% (mainly inactive) Feces: ~40% (mainly unchanged)	Kidney: ~40% (mainly inactive) Feces: ~50% (mainly unchanged)	Kidney: ~57% (mainly inactive) Feces: ~37% (mainly unchanged)

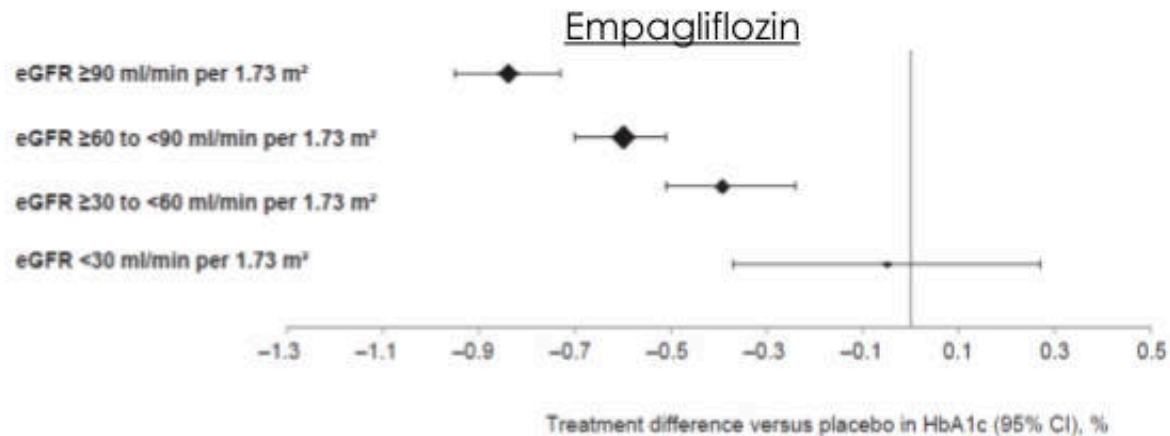
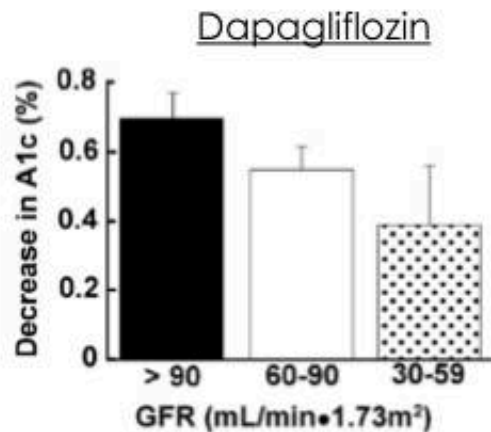


Dose reduction in renal impairment

GFR (mL/min/1.73 m²) Dosage

Canagliflozin	30-60 < 30	- Max dose: ≤ 100 mg/day - Avoid to use
Dapagliflozin	< 45 < 25	- T2DM: Not recommended (HF or CKD: No dosage adjustment required) - Generally not recommended (HF or CKD may continue 10 mg/day)
Empagliflozin	< 30	Generally not recommended (HF or CKD: No dosage adjustment required)
Ertugliflozin	< 45	Avoid to use
Bexagliflozin	< 30	Avoid to use
Sotagliflozin	< 30	Avoid to use

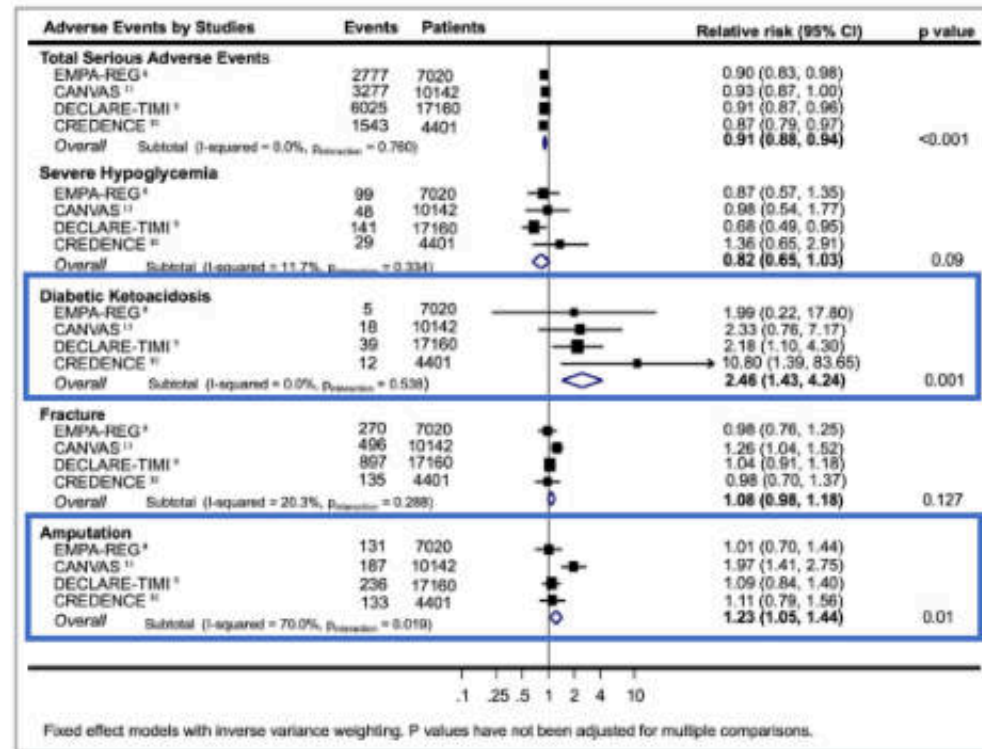
¹Drugs. 2015;75:33–59. ²Clin Pharmacokinet. 2020;59:949–65. ³Invokana®, ⁴Foxiga®, ⁵Jardiance® and ⁶Inpefa® prescribing information. ⁷Am J Kidney Dis. 2018;72(2):267–77. ⁸Drugs. 2023;63:447–53. ⁹Biomedicines. 2023;11:1867. doi: 10.3390/biomedicines11071867. ¹⁰Cardiovasc Diabetol. 2019;18:20. doi: 10.1186/s12933-019-0828.





Physician experiences in Canada

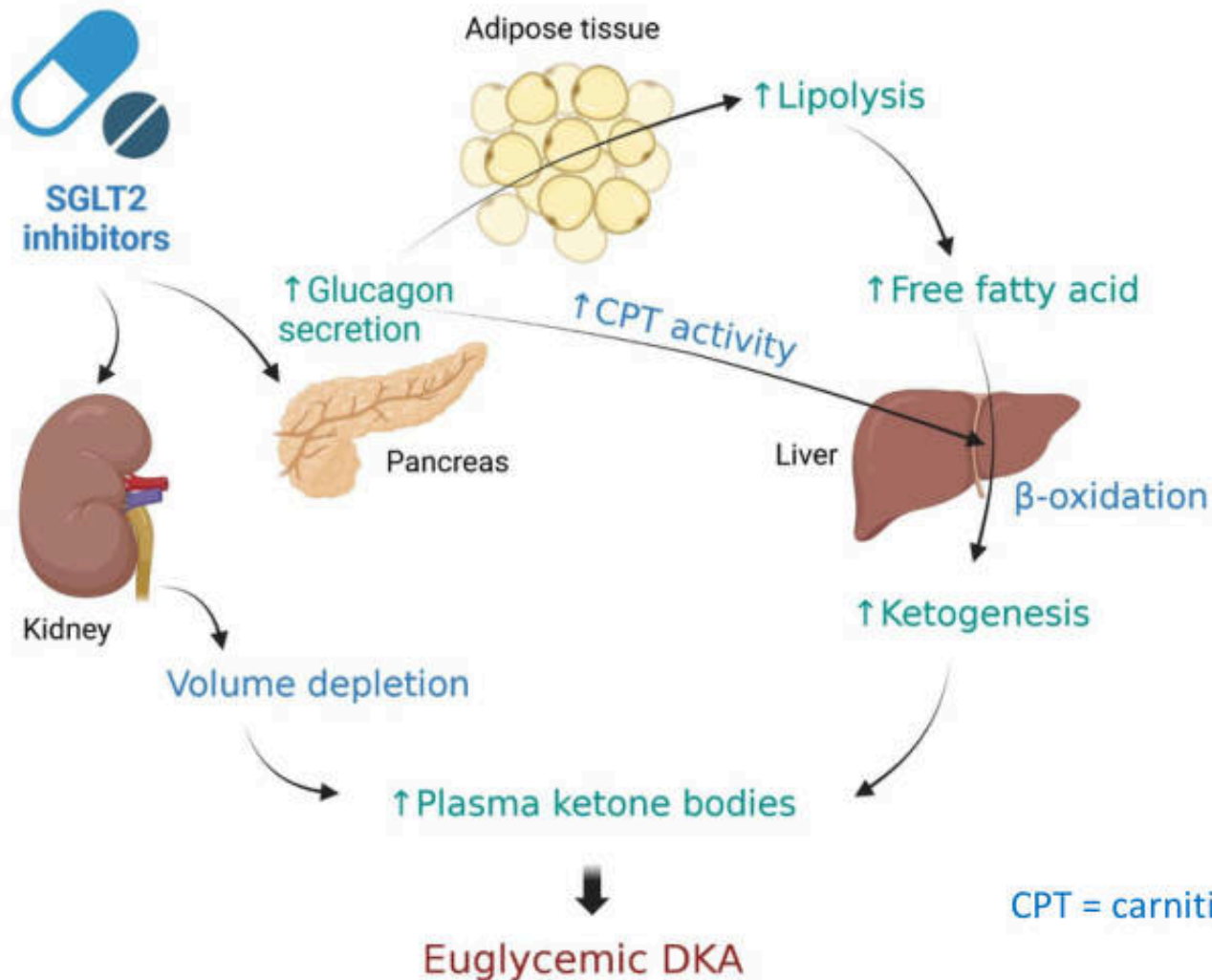
Type of ADR	Number of ADR
• Mycotic infection	82
• DKA	43
• Renal insufficiency	15
• Hyperkalemia	6
Others	
• Urinary infection	7
• Urinary symptoms	6
• Amputation	2
• Severe weight loss	2
• Thirst/dehydration	2



J Am Heart Assoc. 2020;9:e014908.

Level of severity	Complication					
	DKA			Mycotic infection		
	Inpatient	Outpatient	Total	Inpatient	Outpatient	Total
Very mild	0	0	0	3	45	48
Mild	4	0	4	0	28	28
Moderate	28	1	29	3	3	6
Severe	10	0	10	0	0	0

Potential mechanisms whereby SGLT2 inhibitors may increase risk of ketoacidosis





Comparative characteristics of antidiabetic drugs

Drug Class	HbA _{1c} Reduction (%)	Lipids/BP	Body Weight	Major Side Effects
Biguanides	1.0-2.0	↓ TG & LDL ↑ HDL	Neutral/↓	GI disturbances
Sulfonylureas	1.0-2.0	No effect	↑	Hypoglycemia
Non-sulfonylureas	0.5-1.5	No effect	↑	Hypoglycemia
Thiazolidinediones	0.5-2.0	↓ TG ↑ LDL & HDL	↑	Fluid retention , ↓ Hb
GLP-1 receptor agonists	0.5-1.7	↓ TG ↑ HDL	↓	GI disturbances
GIP/GLP-1 receptor agonist	2.0-2.3	↓ TG & LDL ↑ HDL	↓	GI disturbances
DPP-4 inhibitors	0.6-1.5	Neutral	Neutral	Headache, URI
α-glucosidase inhibitors	0.4-1.0	No effect	Neutral	GI disturbances
SGLT2 inhibitors	0.5-1.0	↑ LDL? ↓ BP	↓	Genitourinary infections
Insulin	1.5-3.5	Neutral	↑	Hypoglycemia Lipodystrophy

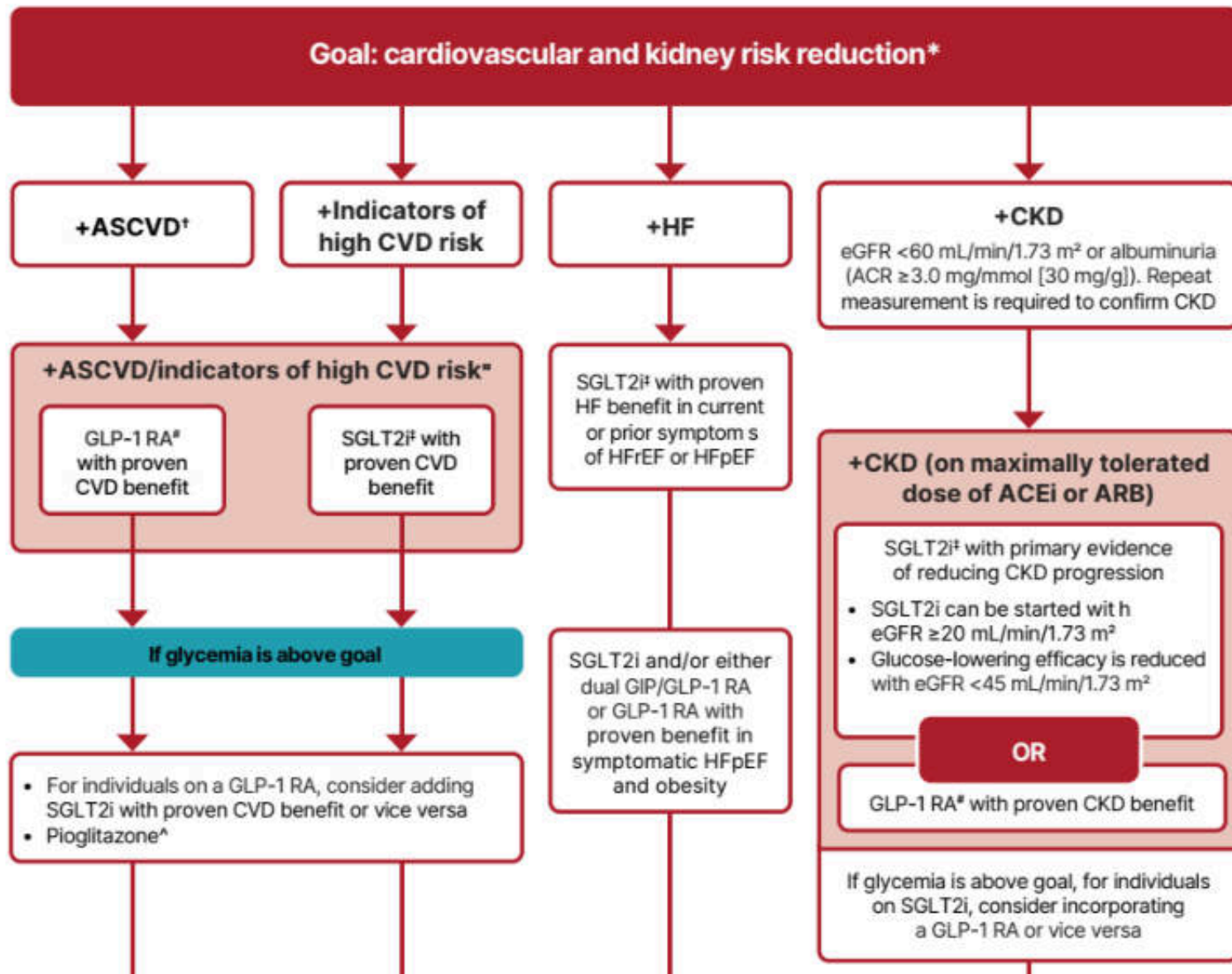


Comparative characteristics of antidiabetic drugs

Drug Class	CV outcomes		
	ASCVD	HF	Renal
Biguanides	Potential benefit	Neutral	Potential benefit
Sulfonylureas	Neutral	Neutral	Neutral
Thiazolidinediones	Potential benefit: Pioglitazone	↑ Risk	Neutral
GLP-1 receptor agonists	Neutral: Lixisenatide Benefit: Liraglutide, Dulaglutide, Semaglutide	Neutral	Benefit: Liraglutide, Dulaglutide, Semaglutide
GIP/GLP-1 receptor agonist	Benefit	Neutral	Neutral
DPP-4 inhibitors	Neutral	Potential risk: Saxagliptin	Neutral
SGLT2 inhibitors	Benefit: Empagliflozin, Canagliflozin Dapagliflozin (in patients with prior MI)	Benefit: Empagliflozin, Canagliflozin, Dapagliflozin, Ertugliflozin Sotagliflozin	Benefit: Empagliflozin, Canagliflozin, Dapagliflozin, Ertugliflozin
Insulin	Neutral	Neutral	Neutral



The ADA Guidelines 2026





The ADA Guidelines 2026



+Mitigating risk of MASLD or MASH



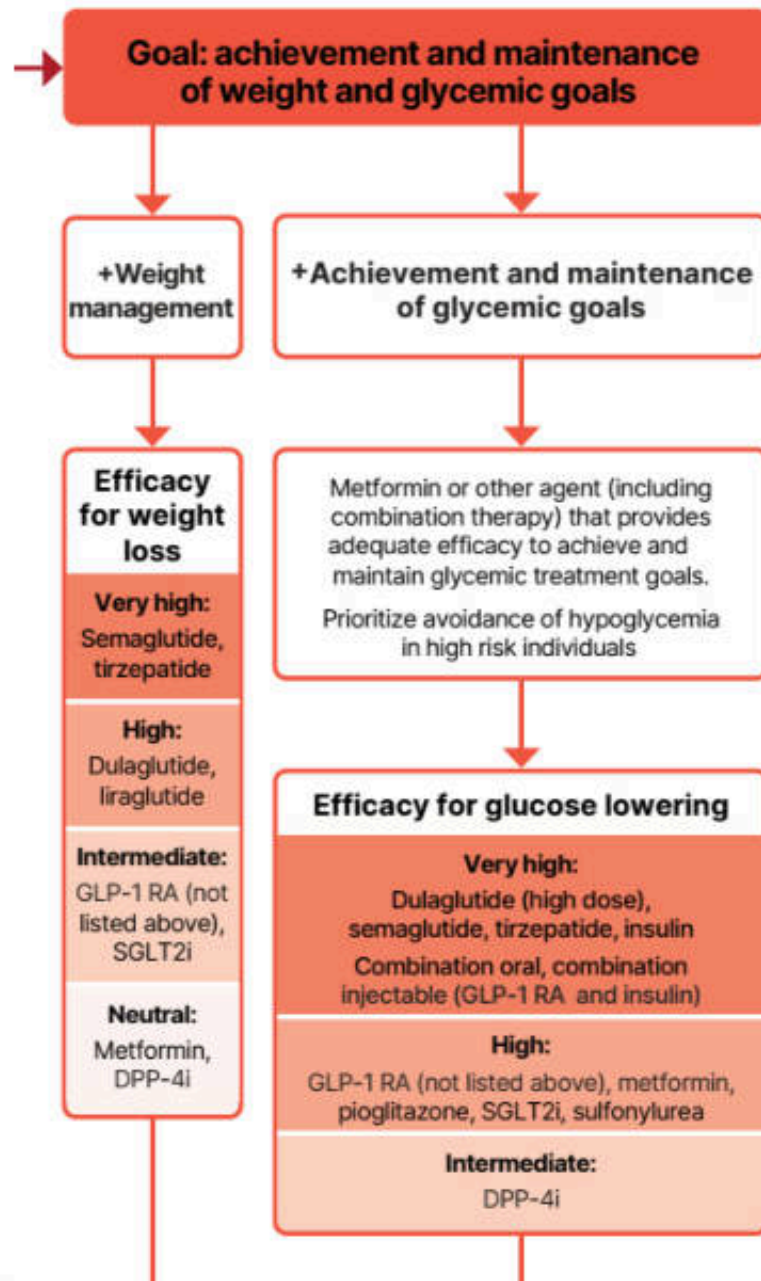
Agents with proven or potential benefit in MASLD or MASH

GLP-1 RA, dual GIP and GLP-1 RA, pioglitazone, or combination of GLP-1 RA with pioglitazone

Use insulin in the setting of decompensated cirrhosis



The ADA Guidelines 2026



Thank You!

